5 PUBLIC LAND SURVEYING SYSTEM

HISTORY OF PUBLIC LANDS

1. The first surveys of public lands in the U.S. were conducted under the provisions of which of the following laws?

(A) the Greenville Treaty
(B) the Act of February 11, 1805
(C) the Virginia Charter
(D) the Land Ordinance Act

Solution: Passed by Congress under the Articles of Confederation in 1785, the Land Ordinance Act established many elements of the Public Land Surveying System that remain virtually unchanged today. The Act created 640-ac sections (which were called *lots*) and 6-mi townships, and required that public lands be surveyed before they were sold.

Answer is (D)

2. Which of the following authorities supervised the first surveys of public lands?

(A) the General Land Office
(B) the Department of the Interior
(C) the Bureau of Land Management
(D) the Geographer of the U.S.

Solution: The General Land Office was proposed by Alexander Hamilton as early as 1790, but was not actually created until 1812. The office of the Surveyor General was first mentioned in the Act of 1796; the Bureau of Land Management did not begin until 1946. The office of the Geographer of the U.S. was created in the Land Ordinance of 1785. Thomas Hutchins was the first and only person to hold the office, and he conducted the first survey of the public lands in the U.S.

Answer is (D)

GLO
1812

3. Rufus Putnam, the Surveyor General in 1797, established the contract system, which continued until 1910. Which statement best characterizes this system?

(A) The contract system provided for credit purchases of public lands by the highest bidders.

(B) The contract system allowed those wishing to purchase land within a township to make a deposit with the government to cover the surveying costs.

(C) The contract system abolished the credit system of purchase of public lands. Under the contract system, buyers were required to pay with hard currency.

(D) The contract system allowed the Surveyors General to arrange contracts with deputy surveyors for the execution of public land surveys.

Solution: From 1797 to 1910, public land surveys and the preparation of field notes were done by surveyors under contract with the General Land Office. These private contractors were generally paid per mile of line surveyed. The contract system was replaced by the direct system, which required that the surveys and resurveys be performed by government employees.

Answer is (D)

4. Thomas Hutchins was the surveyor in charge of the work now known as the Seven Ranges in Ohio, the first public land surveys done under the auspices of the Land Ordinance Act of 1785. He personally ran the line extending east and west from the Ohio River that is known as the Geographer's Line. Lines that serve the same function as the Geographer's Line are now found throughout the Public Land Surveying System. What are such lines called?

(A) sectional guide meridians
(B) guide meridians
(C) sectional correction lines
(D) baselines

Solution: The baselines in the Public Land Surveying System are lines that follow a parallel of latitude from an initial point.

Answer is (D)

5. In 1807, Surveyor General Jared Mansfield ordered that another baseline be established 24 mi from the first in the Vincennes District to compensate for distortions in the survey. In 1818, the same idea was used to compensate for the effect of convergence. Which of the following terms is currently used to refer to such lines?

(A) correction lines
(B) guide meridians
(C) standard parallels
(D) both A and C

Solution: From the beginning, the contradictory rules that townships be both 6-mi square and bounded by cardinal lines plagued the public land surveys. Compensation for convergence of meridians finally was addressed under instructions from Edward Tiffin for the survey in Indiana. The standard parallel, or correction line, was used.

Answer is (D)

6. In 1849, the General Land Office became part of the Department of the Interior. What department of government had jurisdiction over public land surveys before then?

(A) the Department of Commerce
(B) the Department of the Treasury
(C) the Department of Agriculture
(D) the Department of Defense

Solution: One of the initial purposes of the sale of public lands was to raise revenue. After the Revolutionary War, the U.S. had a great deal of land but little money. The sale of public lands provided a method for paying the huge foreign debt, among other things. Therefore, it was natural that the Department of the Treasury was in charge of the early public land surveys.

Answer is (B)

7. The Act of May 10, 1872, sometimes called the Mining Act, established the legal size of a lode claim in the public lands. Which of the following sizes correctly represents this provision of the Mining Act?

(A) 200.00 sq ft
(B) 1320.00 ft by 600.00 ft
(C) 40 chains by 40 chains
(D) 600.00 ft by 1500.00 ft

Solution: Under the 1872 Mining Act, a patented lode claim is 300 ft on each side of the vein for a maximum length of 1500 ft.

Answer is (D)

8. In 1847, Richard Young, commissioner of the General Land Office, instructed Charles Morse in the sectional subdivision of a township that had a particularly crooked eastern boundary. Young's solution was unpopular with Morse, but it limited the alignment problem to only one row of sections by the use of a procedure still used in similar circumstances. What is the procedure?

(A) the establishment of a sectional guide meridian
(B) the establishment of a sectional correction line
(C) subdivision by protraction
(D) a completion survey

Solution: When the alignment of the governing east boundary of a township is irregular, the first meridional line west from the defective boundary is extended north to its intersection with the north boundary of the township. This line is called a *sectional guide meridian*. It allows the subdivision of the sections to its west in accordance with standard practice. East of the sectional guide meridian, fractional portions of the latitudinal section lines are placed in the east half-mile.

Answer is (A)

GUNTER'S CHAIN

9. The chain has been the standard unit of length in the Public Land Surveying System since its beginning and is still the standard prescribed by law. One of the reasons for its durability is its convenient relationship with the description of land in acres. Which of the following equations correctly describes this relationship?

(A) 1 sq chain = 1 ac
(B) 80 sq chains = 40 ac
(C) 10 sq chains = 1 ac
(D) 100 sq chains = 1 ac

Solution: One of the reasons for the eventual rejection of Thomas Jefferson's proposal that the nautical mile be used in the public land surveys was that it was not evenly divisible into chains. The statute mile, equal to 80 chains, was preferred. Under this system, 10 sq chains is equal to 1 ac.

Answer is (C)

10. Which of the following lengths is equal to 0.28 chains?

(A) 19.50 ft
(B) 3 links
(C) 12.00 ft
(D) 28 links

Solution: Gunter's chain is composed of 100 links. One hundredth of a chain is 1 link.

Answer is (D)

11. Early field notes from public land surveys sometimes refer to a two-pole chain. What is a two-pole chain?

(A) A two-pole chain is a chain that is equipped with two long wooden handles.
(B) The phrase two-pole chain is simply another term used to describe a standard-length Gunter's chain.
(C) A two-pole chain is 132.00 ft long.
(D) A two-pole chain is one-half of a standard Gunter's chain, or 33.00 ft long.

Solution: A pole, also known as a *rod* or a *perch*, has a length of 16.5 ft. Therefore, a two-pole chain is a chain that is 33.00 ft in length.

Answer is (D)

12. Gunter's chain was invented by Thomas Gunter in eighteenth century England. The instrument was used in public land surveys throughout the nineteenth century. These chains tended to increase in length with extended use. Why?

(A) the constant repairs required by frequent breaks
(B) the elasticity of the materials from which the chains were made
(C) the wearing down of the links and the rings that joined them together
(D) all of the above

Solution: The earliest Gunter's chains were constructed of iron (later, steel) links joined together by rings. The deterioration of the metal-on-metal joints caused the chains to gradually increase in length.

Answer is (C)

13. Which of the following lengths is equivalent to 79.52 chains?

(A) 1600.00 m
(B) 5248.32 ft
(C) 320.00 rods
(D) 1920.41 varas

Solution: A chain is 66.00 ft.

$$(79.52 \text{ chains})(66.00 \text{ ft}) = 5248.32 \text{ ft}$$

Answer is (B)

THE SOLAR COMPASS

14. William Burt invented his True Meridian Finding Instrument in 1833. Which of the following statements about Burt's invention is correct?

(A) Burt found that the iron ore deposits in Michigan made surveying with a magnetic needle nearly impossible. This led him to invent a solar compass.
(B) Burt's invention provided an instant mechanical solution of the astronomic triangle for the establishment of direction.
(C) Using Burt's invention, the magnetic declination of a compass can be determined.
(D) A, B, and C are true.

Solution: In 1844, the townships of Saginaw Bay were resurveyed under special appropriations from Congress, and the solar compass was the instrument specified for the work. The instrument's faculty for avoiding the hazards of local magnetic attraction made it a valuable tool in areas with high concentrations of mineral deposits.

The 1890 *Manual* outlawed the use of the magnetic needle except in subdivision and meandering; the 1894 *Manual* required that all lines refer to the true meridian.

Answer is (D)

MANUAL OF INSTRUCTIONS

15. The *Manual of Surveying Instructions* for public lands has been published in profoundly different versions periodically since 1855. Which of the following dates corresponds to the version of the *Manual* now in force?

(A) 1947
(B) 1972
(C) 1930
(D) 1973

Solution: The current *Manual* was published in 1973 by the Bureau of Land Management. The previous *Manual* was published in 1947.

Answer is (D)

16. Which of the following statements correctly describes the applicability of the *Manual of Surveying Instructions* with regard to lands that have passed from the public domain into private ownership?

(A) All lands once in the public domain are under the jurisdiction of the Bureau of Land Management.
(B) Lands that have passed from the public domain into private ownership are under the jurisdiction of state law.
(C) Many states have adopted the principles and rules laid down in the *Manual of Surveying Instructions* through state court opinions and statute laws.
(D) Both B and C are true.

Solution: The rules in the 1973 *Manual of Surveying Instructions* are the guidelines by which the Bureau of Land Management must survey public lands today. Original surveys of public lands are presumed to have been in accordance with

the *Manual* or special instructions in force at the time. However, such instructions are only advisory to land surveyors in private practice retracing privately held land, unless the instructions have been adopted by a state statute or state court. Many states have adopted the instructions published by the Bureau of Land Management, making these procedures mandatory in the retracement of public lands within their boundaries.

Answer is (D)

17. Which of the following cannot be found in the 1973 *Manual of Surveying Instructions*?

(A) a summary of the federal statute laws governing the surveying of the public domain
(B) instructions on the adjustment and use of a solar attachment for a transit
(C) rules and procedures for the meandering of swamp and overflowed lands
(D) procedures for the reestablishment of lost corners

Solution: Unlike navigable waters, swamps and overflowed lands are not subject to meandering, so the *Manual* provides no instructions for them.

Answer is (C)

LIMITS AND ADJUSTMENTS

18. Under the 1973 *Manual*, what is the maximum allowable error of closure of a perimeter in either latitude or departure for public land surveys?

(A) 1 part in 5000
(B) 1 part in 2000
(C) 1 part in 10,000
(D) 1 part in 1280

Solution: The error of closure must be calculated after the completion of one or more full or fractional township exteriors, and must not exceed 1 part in 1280.

Answer is (D)

19. A section may be considered regular, that is, not outside the rectangular limits set down in the 1973 *Manual*, if its boundaries do not depart from cardinal directions by more than what prescribed limit?

(A) $00°01'$
(B) $00°10'$
(C) $00°12'$
(D) $00°21'$

Solution: Sectional boundaries cannot be considered regular if they depart by more than $00°21'$ from cardinal directions. This limit, among others, is considered in determining whether lines within a survey are defective.

Answer is (D)

20. The 1973 *Manual* establishes a limit regarding the maximum acceptable adjustment in measurement between regular corners. Which limit must not be exceeded if a section is to be considered regular?

(A) 25 links in 40 chains
(B) 5 links in 80 chains
(C) 10 links in 20 chains
(D) 15 links in 40 chains

Solution: The distance between regular corners in a public lands survey cannot exceed 25 links in 40 chains if the section is to be considered regular.

Answer is (A)

THE QUADRANGLE

21. Which of the following statements is not correct regarding the use of the term *initial point* in the Public Land Surveying System?

(A) Principal meridians and baselines originate at initial points.
(B) The positions of the initial points of the Public Land Surveying System are given in geographical coordinates in the 1973 *Manual*.
(C) Eight public lands surveys have no initial point.
(D) There is an initial point in each of the public lands states.

Solution: The initial point is the origin of the principal meridian and baseline that control the public lands surveys of an area. However, some public lands states have no initial point within their borders (Minnesota, for example), and others have more than one.

Answer is (D)

22. Under current instructions, which of the following types of corners are not to be monumented during the original survey of the principal meridians and baselines?

(A) section corners
(B) quarter-section corners
(C) meander corners
(D) closing corners

Solution: The procedure for running the principal meridians and baselines of the Public Land Surveying System calls for the monumentation of all quarter-section, section, and township corners along the lines. Meander corners are to be monumented as well when these standard lines intersect meanderable bodies of water. Closing corners are set during subsequent surveys.

Answer is (D)

23. The standard lines of the Public Land Surveying System include principal meridians, baselines, standard parallels, and guide meridians. Which of the following statements is not true regarding the general plan for the establishment of these standard lines under current instructions?

(A) Guide meridians and principal meridians are intended to follow astronomical meridians within 00°03′.

(B) Standard parallels and baselines are intended to follow parallels of latitude within 00°03′.

(C) The difference between two sets of measurements of a standard line should not differ by more than 25 links in 40 chains.

(D) Double corners will eventually stand on standard parallels and baselines.

Solution: The maximum allowable discrepancy in measurement of rectangular limits in subdivisional surveys is 25 links in 40 chains, but the survey of the standard lines must be run to a higher precision. The allowable discrepancy between the two sets of measurements along a standard line is 7 links per 80 chains.

Answer is (C)

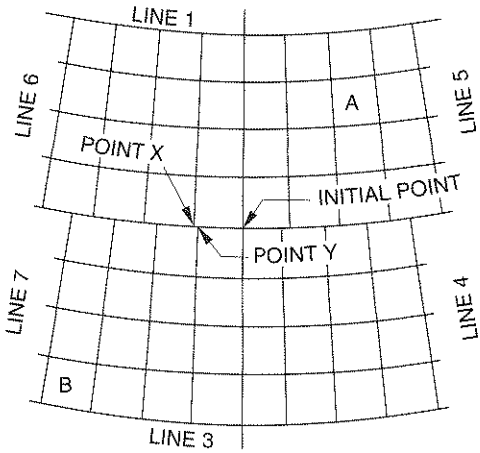
24. In the past, guide meridians and standard parallels were not always established at intervals of 24 mi. Which other interval would be most likely?

(A) 23 mi
(B) 33 mi
(C) 21 mi
(D) 36 mi

Solution: Townships are intended to be 6-mi square, so the interval between successive guide meridians or standard parallels would most likely be divisible by six.

Answer is (D)

The following five problems refer to the illustration shown.



25. Which of the following labels correctly identify lines 4 and 5?

(A) Line 4 is the First Guide Meridian East. Line 5 is the Second Guide Meridian East.

(B) Line 4 is the First Correction Line East. Line 5 is the Second Correction Line East.

(C) Line 4 is the First Standard Parallel. Line 5 is the Second Standard Parallel.

(D) Lines 4 and 5 are portions of the First Guide Meridian East.

Solution: The guide meridians are numbered consecutively east and west from the principal meridian. They follow true astronomical meridians, so they converge.

Answer is (D)

26. Which of the following labels correctly identify lines 1 and 3?

(A) Line 1 is the First Standard Parallel North. Line 3 is the First Standard Parallel South.

(B) Line 1 is the First Correction Line North. Line 3 is the First Correction Line South.

(C) Line 1 is the First Latitudinal Section Line North. Line 3 is the First Latitudinal Section Line South.

(D) Both A and B are true.

Solution: The standard parallels are numbered consecutively north and south from the baseline. Standard parallels are also known as *correction lines*.

Answer is (D)

27. Which of the following labels correctly identify points X and Y?

(A) Point X is a standard corner. Point Y is a closing corner.

(B) Point X is a meander corner. Point Y is a witness corner.

(C) Point X is a standard corner. Point Y is a witness corner.

(D) Point X is a closing corner. Point Y is a standard corner.

Solution: Point X is a standard corner and would be set 6 mi west from the initial point in an idealized quadrangle. Due to the convergence of meridians, point Y would be set less than 6 mi from the initial point. It is known as a *closing corner*.

Answer is (A)

28. Which of the following labels would correctly identify the township labeled A?

(A) T2N R3E
(B) T3N R3W
(C) T3S R2W
(D) T3N R3E

Solution: Ranges are numbered consecutively east and west from the principal meridian. Townships are numbered consecutively north and south from the baseline. This system continues throughout the area covered by a particular principal meridian and baseline.

Answer is (D)

29. Which of the following statements is not true concerning the boundaries of township B under current instructions?

(A) The eastern township boundary should be run parallel with the principal meridian.
(B) The northern township boundary should be run on a random line and returned on a true line.
(C) As the township boundaries are run, quarter-section and section corners should be set alternatively at intervals of 40 chains.
(D) The eastern and western township boundaries should be run from the south to the north.

Solution: The meridional township boundaries are intended to follow true astronomical meridians, not to be parallel with the principal meridian.

Answer is (A)

30. Township corners are normally established at intervals of 480 chains. However, current instructions specify a somewhat different procedure regarding the original closing of meridional township boundaries on standard parallels. Which of the following statements correctly characterizes those instructions?

(A) The excess or deficiency of measurement must be placed in the last quarter mile, between the last set quarter-quarter corner monument and the closing corner.

(B) A closing township corner must be established 1920 chains from the standard parallel to the south.

(C) The meridional township boundary must be ended at its intersection with the standard parallel, and the excess or deficiency of measurement placed in the last half mile.

(D) Meridional Township boundaries run from north to south.

Solution: The closing of a meridional township boundary on a standard parallel is similar to the closing of a guide meridian on a standard parallel. In both cases, a closing corner is set at the intersection of the meridian with the parallel. The meridional township boundaries are begun at the standard corners set at intervals of 480 chains along the standard parallel on the southern boundary of a quadrangle. The boundaries end at the closing corners, which, due to the convergence of the meridians, will be closer together than 480 chains along the standard parallel, which is the northern boundary of a quadrangle.

Answer is (C)

THE TOWNSHIP

The following three problems refer to the illustration shown.

6	5	4	3	2	1
7	8	9	10	11	12
18	17	16	15	14	13
19	20	21	22	23	24
30	29	28	27	26	25
31	32	33	34	35	36

N

T4N R3E

31. Under normal conditions and by current instructions, which section's boundaries are established first in the subdivision of a regular township?

(A) Section 1
(B) Section 6
(C) Section 31
(D) Section 36

Solution: The most easterly range of sections is the first to be surveyed in the subdivision of a regular township under current instructions. The subdivisional survey begins at the SW corner of Section 36. The western boundary of Section 36 is established first, then the survey proceeds across its northern boundary and continues along the western boundary of Section 25.

Answer is (D)

32. Under normal conditions and by current instructions, which of the followir statements correctly describes the alignment of the eastern and northern bounc aries of Section 22?

(A) The eastern boundary is intended to follow a true astronomical meridia of longitude. The northern boundary is intended to follow a parallel (latitude.

(B) The eastern boundary is intended to be parallel with the eastern boundar of the township. The northern boundary is intended to be parallel wit the southern boundary of the township.

(C) The eastern boundary is intended to be parallel with the western bounc ary of the township. The northern boundary is intended to be paralle with the northern boundary of the township.

(D) The eastern boundary is intended to follow a geodetic meridian of long tude. The northern boundary is intended to follow a geodetic parallel c latitude.

Solution: The eastern and southern boundaries of a township are often know as the *governing boundaries*. The meridional section lines in a township ar intended to be parallel with the eastern boundary of the township, while th latitudinal boundaries are intended to be parallel with the southern boundary c the township.

Answer is (B)

33. Referring to the illustration of T4N R3E, which of the following statement is correct regarding the typical placement of closing corners in such a township'

(A) Closing corners would normally occur along its eastern boundary.

(B) Closing corners would normally occur along its northern boundary.

(C) There would normally be no closing corners in such a township.

(D) Closing corners would normally occur along its western boundary.

Solution: The northern boundary of T4N R3E would normally be a standarc parallel. The meridional section lines along the northern tier would, under typica conditions, terminate in closing corners.

Answer is (B)

34. Under normal conditions and by current instructions, which of the following statements is not true regarding the establishment of the quarter-section corners during the subdivisional survey of a township?

(A) Quarter-section corners are normally established on the line halfway between section corners along the section lines of the 25 regular sections in a township.

(B) Quarter-section corners are usually established alternatively with section corners at intervals of 40 chains along the section lines of the 25 regular sections in a township.

(C) Temporary quarter-section corners are established during the running of a random line, and are corrected to their proper position during the running of the true line along each section line.

(D) Quarter-section corners are established along the northern tier of sections, 40 chains north of the standard section corner, leaving the excess or deficiency of measurement in the last half mile.

Solution: Temporary quarter-section corners are not set on the meridional section lines since these lines are not established using the random and true line method, except when closing on a standard parallel. Most meridional section lines are run on the true line, unlike the latitudinal section lines, which are most often established using a random and true line method.

Answer is (C)

35. Which of the following correctly defines the term *fractional township*?

(A) a township that has been subdivided by protraction

(B) a township that has not been fully subdivided

(C) a township that contains significantly fewer than 36 sections, usually due to an invasion by a segregated body of water

(D) a township that contains fractional sections

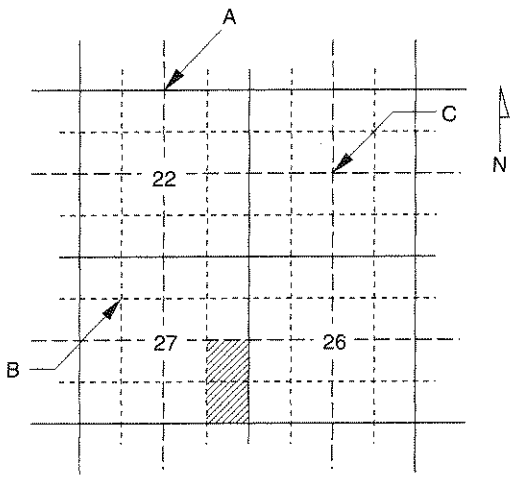
Solution: A township that cannot be normally subdivided into its full complement of 36 sections due to the presence of water—which cannot be properly included within the township or closings—is considered a fractional township.

Closings on state boundaries or conditions that cause the creation of half-ranges or half-townships are typical causes of fractional townships.

Answer is (C)

THE SECTION

The following seven problems refer to the illustration shown.



36. Which name is correct for the corner labeled A?

(A) an aliquot corner
(B) a center sixteenth corner
(C) the S$\frac{1}{4}$ corner of Section 15
(D) both A and C

Solution: An aliquot corner in the Public Land Surveying System is a corner of an aliquot part. Aliquot parts are established by the legal subdivision of sections into halves and fourths. These halves and fourths may then be subdivided into further aliquot parts, ad infinitum.

The corner labeled A is an aliquot corner. It is also the S$\frac{1}{4}$ section corner of Section 15. Section 15 is directly north of Section 22 in a regular township.

Answer is (D)

37. Which name is correct for the corner labeled B?

(A) the NW$\frac{1}{16}$ corner of Section 27
(B) the NW corner of the SE$\frac{1}{4}$ of the NW$\frac{1}{4}$ of Section 27
(C) the NE corner SW$\frac{1}{4}$NW$\frac{1}{4}$ of Section 27
(D) all of the above

Solution: The corner labeled B is described by all of the answers given. Other names would also be correct, including the SW corner of the NE$\frac{1}{4}$NW$\frac{1}{4}$ of Section 27.

Answer is (D)

38. Which name is correct for the corner labeled C?

(A) the C$\frac{1}{4}$ of Section 21
(B) the NE corner of the SW$\frac{1}{4}$SW$\frac{1}{4}$ of Section 23
(C) the SW corner of the SE$\frac{1}{4}$NE$\frac{1}{4}$ of Section 23
(D) the NE$\frac{1}{16}$ of Section 21

Solution: The corner labeled C in Section 23 is not correctly described by any of the names given. One name that would correctly describe it is the center quarter (C$\frac{1}{4}$) of Section 23.

Answer is (D)

39. Which of the labeled corners is located by the intersection of lines rather than by measurement along one line?

(A) corner A
(B) corners A and C
(C) corners C and B
(D) corners A and B

Solution: The center quarter corners and the center sixteenth corners of sections are located by intersection of quarter lines and sixteenth lines, respectively. The N$\frac{1}{4}$ corner of Section 22, like an aliquot corner along a sectional boundary, is located by measurement along the section line.

Answer is (C)

40. Under current instructions, which of the labeled corners would usually be monumented during the subdivisional survey of a township?

(A) corner A
(B) corner B
(C) corner C
(D) all of the above

Solution: Quarter-section corners that occur on the section lines within a township are usually monumented during the subdivision of a township into sections. The sixteenth and center quarter-section corners may occasionally be set under special instructions.

Answer is (A)

41. If Section 27 is a regular section containing 640 ac, what is the acreage of its crosshatched aliquot part?

(A) 10 ac
(B) 60 ac
(C) 40 ac
(D) 80 ac

Solution: The crosshatched area is approximately 40 chains by 20 chains. The calculation of the acreage of the crosshatched area is made more convenient by remembering that 10 sq chains are equal to 1 ac.

$$(40 \text{ chains})(20 \text{ chains}) = 800 \text{ sq chains}$$

$$(800 \text{ sq chains})\left(\frac{1 \text{ ac}}{10 \text{ sq chains}}\right) = 80 \text{ ac}$$

Answer is (D)

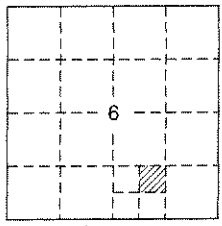
42. Which of the following phrases best describes the crosshatched aliquot part of Section 27?

(A) the E$\frac{1}{4}$ of the S$\frac{1}{2}$ of Section 27
(B) the SE$\frac{1}{4}$ of the E$\frac{1}{2}$ of Section 27
(C) the E$\frac{1}{2}$ of the SE$\frac{1}{4}$ of Section 27
(D) the E$\frac{1}{4}$ of the SE$\frac{1}{2}$ of Section 27

Solution: The crosshatched area is in the SE$\frac{1}{4}$ of Section 27, and is its E$\frac{1}{2}$.

Answer is (C)

The following three problems refer to the illustration shown.



43. If the SE$\frac{1}{4}$ of Section 6 is regular and contains 160 ac, what is the acreag of the crosshatched aliquot part?

(A) 40 ac
(B) 10 ac
(C) 5 ac
(D) 20 ac

Solution: The crosshatched area is approximately 10 chains by 10 chains.

$$(10 \text{ chains})(10 \text{ chains}) = 100 \text{ sq chains}$$

$$(100 \text{ sq chains})\left(\frac{1 \text{ ac}}{10 \text{ sq chains}}\right) = 10 \text{ ac}$$

Answer is (B)

44. Which of the following phrases best describes the crosshatched aliquot par in Section 6?

(A) the NE$\frac{1}{4}$ of the SW$\frac{1}{4}$ of the SE$\frac{1}{4}$ of Section 6
(B) the NE$\frac{1}{4}$ of the SE$\frac{1}{4}$ of the SW$\frac{1}{4}$ of Section 6
(C) the NE$\frac{1}{4}$ of the SE$\frac{1}{2}$ of the SE$\frac{1}{4}$ of Section 6
(D) the NE$\frac{1}{4}$ of the NE$\frac{1}{4}$ of the SE$\frac{1}{4}$ of Section 6

Solution: The crosshatched aliquot part in Section 6 is in the SE$\frac{1}{4}$ of the section. It is in the SW$\frac{1}{4}$ of the SE$\frac{1}{4}$ of the section. Finally, it is the NE$\frac{1}{4}$ of the SW$\frac{1}{4}$ of the SE$\frac{1}{4}$ of the section.

Answer is (A)

45. Which of the following names would correctly apply to the crosshatched aliquot part in Section 6?

(A) quarter-quarter
(B) sixteenth
(C) two hundred fifty-sixth
(D) sixty-fourth

Solution: The names given to aliquot parts of a section are based on the ratio of their area to the area of the full section. For example, a quarter section would be approximately one-quarter of the area of the full section, or 160 ac.

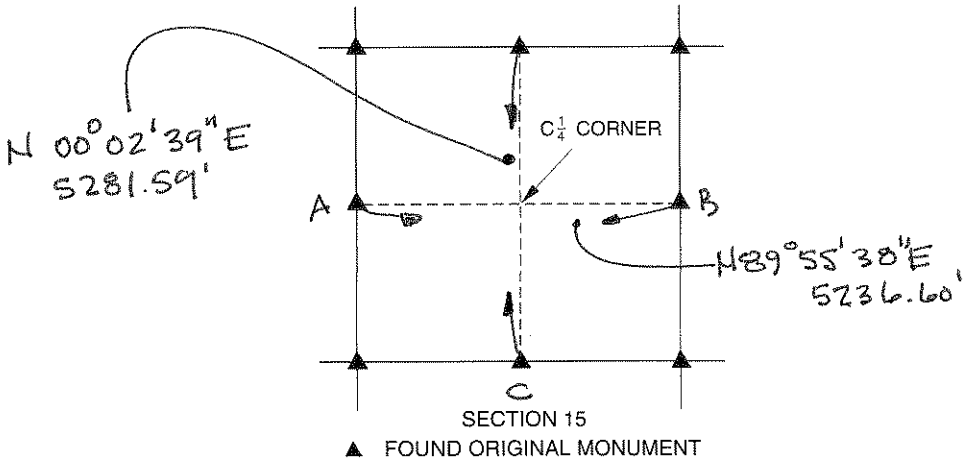
The crosshatched portion of Section 6 contains approximately 10 ac.

$$\frac{\text{approximate area of the aliquot part}}{\text{approximate area of the full section}} = \frac{10 \text{ ac}}{640 \text{ ac}} = \frac{1}{64}$$

The aliquot part crosshatched in Section 6 is a sixty-fourth part of the area of the full section. It is called a sixty-fourth.

Answer is (D)

46. Referring to the illustration and coordinates given (which are based on field measurements taken between the found corner monuments), what are the coordinates of the center quarter corner of Section 15?



D	N¼ corner:	N10,280.3212 E7618.1717	NE corner:	N10,287.9181 E10,229.7806
B	E¼ corner:	N7642.6827 E10,241.1944	SE corner:	N4997.4476 E10,252.6029
C	S¼ corner:	N4998.7326 E7614.0897	SW corner:	N5000.0000 E5000.0000
A	W¼ corner:	N7636.0259 E5004.6007	NW corner:	N10,272.7320 E5009.2027

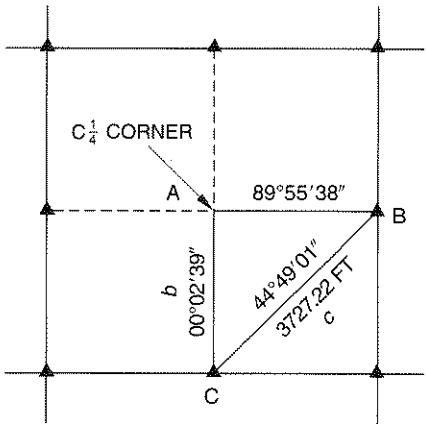
(A) N7639.3431
E7616.1252

(B) N7639.3543
E7622.8976

(C) N7639.5269
E7616.0965

(D) N7634.9800
E7622.7431

Solution: The straight lines that connect the quarter-section corners on opposite sides of the section intersect at the center quarter corner. The point of intersection can be readily found by using a programmable calculator, or by solving the triangle created by the E$\frac{1}{4}$, S$\frac{1}{4}$, and C$\frac{1}{4}$ corners.



SECTION 15

▲ FOUND ORIGINAL MONUMENT

The azimuths and distance shown are found by inversing between the given coordinates. The triangle may be solved by the law of sines.

$$\mathrm{A} = 180^\circ - 89^\circ 55' 38'' + 0^\circ 02' 39'' = 90^\circ 07' 01''$$
$$\mathrm{B} = 89^\circ 55' 38'' - 44^\circ 49' 01'' = 45^\circ 06' 37''$$
$$\mathrm{C} = 44^\circ 49' 01'' - 0^\circ 02' 39'' = 44^\circ 46' 22''$$
$$\frac{a}{\sin \mathrm{A}} = \frac{b}{\sin \mathrm{B}} = \frac{c}{\sin \mathrm{C}}$$
$$\frac{3727.22 \text{ ft}}{\sin 90^\circ 07' 01''} = \frac{b}{\sin 45^\circ 06' 37''}$$
$$b = 2640.62 \text{ ft}$$
$$\frac{3727.22 \text{ ft}}{\sin 90^\circ 07' 01''} = \frac{c}{\sin 44^\circ 46' 22''}$$
$$c = 2625.08 \text{ ft}$$

These distances and the azimuths found by inversing make it possible to find tl coordinates of the C$\frac{1}{4}$ corner.

C$\frac{1}{4}$ corner: N7639.3431
E7616.1252

Answer is (A)

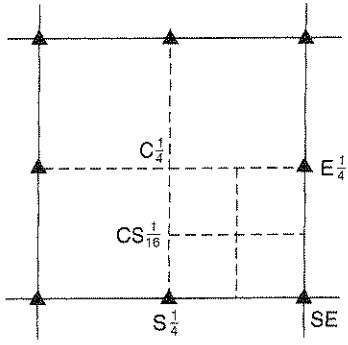
47. Which of the following statements is true regarding the center quarter corne of a regular section under normal circumstances?

(A) The center quarter corner does not exist until it is monumented.
(B) The center quarter corner stands at the midpoint of the north-sout (meridional) centerline of a section.
(C) The center quarter corner stands at the midpoint of the east-west (lat tudinal) centerline of a section.
(D) The center quarter corner almost certainly does not stand at the midpoin of any aliquot line.

Solution: The center quarter corner is found by intersecting the centerline through a regular section. The intersection of the two lines almost certainl does not bisect either line.

Answer is (D)

48. Which of the following statements correctly describes the position of th CS$\frac{1}{16}$ corner in the illustration?



SECTION 15
▲ FOUND ORIGINAL MONUMENT

(A) The quarter-quarter corner labeled CS$\frac{1}{16}$ should be placed 20 chains north from the S$\frac{1}{4}$ corner.

(B) The $\frac{1}{16}$ corner labeled CS$\frac{1}{16}$ should be placed 20 chains south from the center quarter corner.

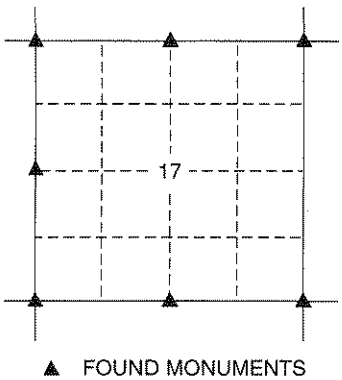
(C) The CS$\frac{1}{16}$ corner should be placed at the intersection of the first sixteenth line north of the south section line with the meridional quarter line.

(D) The CS$\frac{1}{16}$ corner should be placed at the midpoint of the line between the C$\frac{1}{4}$ and the S$\frac{1}{4}$ of the section.

Solution: The quarter-quarter corners, or sixteenth corners, of a section are normally placed at the midpoint of the lines between established corners. The exceptions to such placement usually involve closing corners, which normally occur in the sections along the northern and western boundaries of a township.

Answer is (D)

49. How many public land corners are indicated in the illustration?

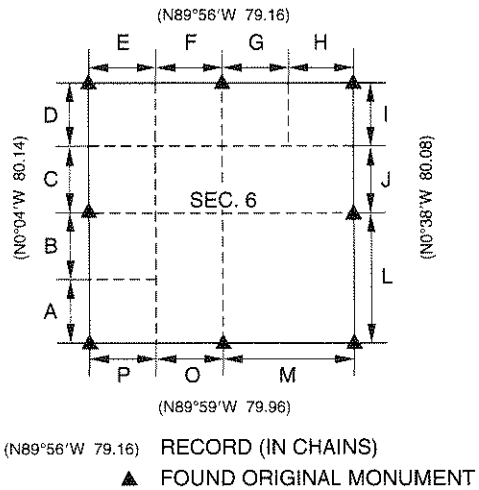


(A) 7
(B) 8
(C) 9
(D) 25

Solution: Each of the quarter-quarter corners, the quarter corners, and the section corners is considered a public land corner. Even though all 25 corners are not monumented, they are nevertheless corners.

Answer is (D)

The following five problems refer to the illustration shown.



Section 6 is part of a township where the normal closing distances occur on the northern and western boundaries of the township.

50. The broken lines shown in the illustration are typical of those shown in closing sections on official township plats. Which of the following statements is correct regarding the broken lines?

(A) They are lines that were normally measured during the original survey of the township.

(B) The broken lines in Section 6 indicate surveyed lines dividing the section into two half-quarter sections, one quarter section, and eight quarter-quarter sections.

(C) The broken lines are created by protraction.

(D) The broken lines divide Section 6 into two half-quarter sections, one quarter section, and eight lots.

Solution: The broken lines found on official township plats subdivide the sections by protraction. These lines exist only on paper and were not normally surveyed in the field.

Answer is (C)

51. What is the measurement implied by the information given in the illustration for the distance labeled E?

(A) 20.08 chains
(B) 19.96 chains
(C) 20.11 chains
(D) 19.16 chains

Solution: The excess or deficiency of the original subdivision along the latitudinal boundaries of the township is absorbed by the last quarter mile. The last half mile is not subdivided at its midpoint. The sixteenth corner is set 20 chains west from the last regular quarter section corner, throwing the excess or deficiency into the last quarter mile.

Answer is (D)

52. What is the measurement implied by the information given in the illustration for the distance labeled L?

(A) 40.00 chains
(B) 40.04 chains
(C) 40.07 chains
(D) 39.97 chains

Solution: One of the principles of the subdivision of sections within a township is to provide as many regular aliquot parts as possible. Throwing the excess or deficiency into the northern and western sides of the township allows the remainder of the subdivisions to be regular. This principle was first established by the Act of May 10, 1800.

Answer is (A)

53. Which of the pairs contains the label of a distance that is not intended to be 20.00 chains on the official plat?

(A) H and I
(B) C and B
(C) B and F
(D) G and O

Solution: The only distance that closes on the north or west boundary of the township is I. The distance labeled I should be 20.08 chains rather than 20.00 chains.

Answer is (A)

54. A quarter-quarter corner is also known as which of the following?

(A) a sixteenth corner
(B) a sixty-fourth corner
(C) a two hundred fifty-sixth corner
(D) a meander corner

Solution: In the Public Land Surveying System, the term *aliquot* means the division of a section or other aliquot part by one-half or one-quarter. This division is not precise in terms of area, but is based upon the bisection and intersection of the lines bounding other aliquot parts. A section contains approximately 640 ac, a quarter section contains approximately 160 ac, and a quarter-quarter ($\frac{1}{16}$) of a section contains approximately 40 ac. Therefore, the corners of the quarter-quarter of a section are sometimes called sixteenth corners.

Answer is (A)

55. Which of the following aliquot parts would not be found in Section 6?

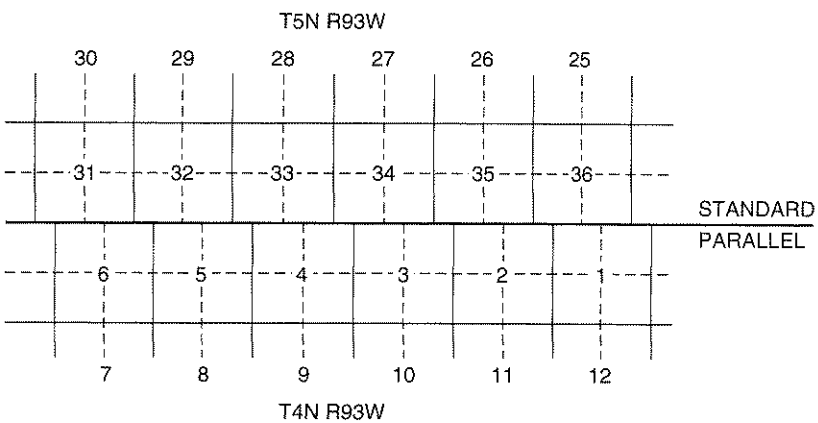
(A) one quarter section
(B) two half-quarter sections
(C) one quarter-quarter section
(D) eight quarter-quarter sections

Solution: Since they absorb any excess or deficiency, the portions of Section 6 protracted along the north and west boundaries are irregular. They are called *lots*, not aliquot parts.

Answer is (D)

CLOSING CORNERS

The following three problems refer to the illustration shown, which shows a standard parallel under usual conditions surveyed by current instructions.



56. Which of the corners along the standard parallel would be called closing corners, and which would be called standard corners?

(A) All of the corners would be known as closing corners.
(B) All of the corners would be known as standard corners.
(C) The corners of Sections 31 through 36 would be known as standard corners, and the corners of Sections 1 through 6 would be known as closing corners.
(D) The corners of Sections 31 through 36 would be known as closing corners, and the corners of Sections 1 through 6 would be known as standard corners.

Solution: Township and section lines are closed on standard parallels, also known as *correction lines*. Closing corners are normally established at the intersection of these closing lines and the correction lines. Therefore, the corners of the sections that are south of the correction line are known as *closing corners*, and the corners of the sections that are north of the correction line are known as *standard corners*.

Answer is (C)

57. In what order were the corners shown established by the original surveyors?

(A) First: the township, section, and quarter-section corners for T5N R93W
Second: the township corners for T4N R93W
Third: the closing section and quarter-section corners for T4N R93W
(B) First: the township corners for both T5N R93W and T4N R93W
Second: the section and quarter-section corners for T5N R93W
Third: the closing section and quarter-section corners for T4N R93W
(C) First: the township, section, and quarter-section corners for T5N R93W
Second: the closing section and quarter-section corners for T4N R93W
Third: the township corners for T4N R93W
(D) First: the township corners for T4N R93W
Second: the township, section, and quarter-section corners for T5N R93V
Third: the closing section and quarter-section corners for T4N R93W

Solution: Under normal circumstances, the first corners established on a standard parallel are the standard township, section, and quarter-section corners.

These corners are set during the initial running of the standard parallel. When the quadrangle defined by the standard parallels and guide meridians is divided into townships, the township lines from south of the standard parallel are closed upon it, and closing corners are set at the intersections. Finally, when the township south of the standard parallel is subdivided into sections, the section lines are closed upon the standard parallel and closing corners are set at the intersections.

Answer is (A)

58. A monument said to be a closing corner is found near the intersection of the west line of Section 1 and the standard parallel. The monument appears to have been set by the original surveyor of T4N R93W. However, it is 12.00 ft north of the standard parallel. Which of the following statements describes the best course of action under these circumstances?

(A) If set by the original surveyor, the monument is indeed the closing corner.
(B) The monument should be destroyed and the proper closing corner set at the intersection of the west line of Section 1 and the standard parallel.
(C) If set by the original surveyor, the monument should be used to govern the direction of the west line of Section 1, but not the line's termination.
(D) Both A and C are best.

Solution: Original monuments should not be destroyed. A closing corner monument found off the standard parallel does determine the direction of the closing line, in this case the west line of Section 1. However, while the monument may not stand on the standard parallel, the closing corner must. The monument controls the direction of the closing line, but not its terminus. The closing corner is at the intersection.

Answer is (C)

59. Which of the following terms would correctly complete this excerpt from the original field notes?

> 74.84 Intersected Base Line at a point 21 links East of Standard, at which point I set a post in mound with a charred stick for a
>
> _____ _____
>
> to Sections 4 & 5.

(A) township corner
(B) meander corner
(C) witness corner
(D) closing corner

Solution: The baseline referred to in the field notes is a standard parallel or correction line; standard refers to a standard corner that had been previously established when the baseline was surveyed. The line between Sections 4 and 5 would be a closing line in such a situation, so the missing words are: closing corner.

Answer is (D)

60. Closing corners are sometimes set where a surveyed line intersects a previously established boundary. Which of the following situations would not cause an original surveyor of the public domain to set a closing corner?

(A) the intersection of a section line with the boundary of a previously established Spanish land grant
(B) the intersection of a township line with a meander line previously surveyed around a navigable river
(C) in a resurvey, the intersection of a section line with a sectional guide meridian
(D) the intersection of a township boundary with the boundary of a state

Solution: Once a boundary is established and bona fide rights have arisen, those rights cannot be violated by subsequent surveys of the public lands. This principle guides the establishment of closing corners in situations where previously settled boundaries would otherwise be disturbed.

In the case of the intersection of a standard, section, or township line with a meander line, no closing corner is set. A meander corner is established, but such a corner is distinctly different from a closing corner. A meander line is not intended to represent a legal boundary, but rather to indicate the segregation of a body of water to ascertain the quantity that remains. Therefore, the meander corner, unlike a closing corner, is not intended to be a legal boundary monument.

Answer is (B)

61. Which of the following markings would a surveyor not expect to find on the brass tablet of a monument at a closing corner along the boundary between two states?

(A) the abbreviation CC
(B) the year the monument was set
(C) the names of the adjoining states
(D) the names of the adjoining townships, ranges, and sections

Solution: Corners defining boundaries between states are not set under general authority of the Bureau of Land Management. They were occasionally set under specific authorization of Congress by the General Land Office, and sometimes are resurveyed under the direction of the Supreme Court or the states, with Congress' consent. Closing corners set during the normal couse of a survey of the public domain are not stamped with the names of the states. The absence of such stamping is intended to prevent the closing corners from being misconstrued as definitions of the state boundaries.

Answer is (C)

62. During the original subdivision of a township into sections, under current instructions, which of the following statements is correct concerning closing corners?

(A) The northern township boundary must always include closing corners.
(B) The section lines that close upon the northern and western township boundaries are always surveyed by the random and true line procedure.
(C) Following the establishment of a closing corner upon a standard parallel, the surveyor should retrace the line closed upon to the east and west of the intersection.
(D) Current instructions prohibit the establishment of closing corners on the southern and eastern boundaries of townships.

Solution: Careful measurement of the distance and direction from a closing corner monument to the nearest standard to the east and west corners is suggested as a check on the closing corner's location. This procedure also facilitates the reestablishment of a lost closing corner.

Answer is (C)

CORNERS, MONUMENTS, AND ACCESSORIES

63. The principle behind the use of double proportionate measurement in the restoration of lost corners in the public lands system is best defined by which of the following statements?

(A) Monuments and corners are the same.
(B) Distance is superior to direction.
(C) Original measurements are nearly always incorrect.
(D) Modern surveying instruments can correct mistakes of the past.

Solution: Directions are ignored in double proportionate measurement in the Public Land Surveying System. This method of reestablishment of lost monuments is generally applied to lost corners common to four townships and four interior sections.

Answer is (B)

64. Corner accessories include which of the following?

(A) line trees
(B) bearing trees and bearing objects
(C) pits and memorials
(D) both B and C

Solution: Corner accessories are considered part of the monumentation of the corner itself. They are physical objects whose distances and/or directions from the corner are known.

Answer is (D)

65. Which of the following best describes a corner for which no evidence of any kind can be found?

(A) an obliterated corner
(B) a lost corner
(C) an extant corner
(D) an existent corner

Solution: In the Public Land Surveying System, three terms are used to describe the condition of monumentation: existent, obliterated, and lost. An existent corner can be found by its monumentation, accessories, field notes, or some other acceptable supplemental evidence or testimony. An obliterated corner cannot be located by evidence of its monumentation or accessories, but may be located beyond a reasonable doubt by testimony or some acceptable record evidence. A lost corner cannot be located by testimony, physical, or record evidence. A lost corner can only be reestablished by reference to other known corners.

Answer is (B)

66. Which of the following statements best defines the difference between a corner and a monument?

(A) Corner and monument are synonymous terms.
(B) A monument that can be verified as having been set by the original surveyor in a public lands survey is exactly at the position of the corner it purports to be, except off-line closing corner monuments.
(C) A corner is a point where boundaries meet, while a monument is physical evidence of a corner's location on the ground.
(D) Both B and C are true.

Solution: A corner is a point on the earth's surface where two lines meet, and a monument is a physical object that marks the position of a corner. While the terms are frequently used synonymously, strictly speaking they are not the same. In the Public Land Surveying System, a monument set by an original surveyor to represent a particular corner is understood to exactly locate the position of that corner, except off-line closing corner monuments. However, subsequent monuments may or may not stand precisely in the position of the corners they are intended to represent.

Answer is (D)

67. Should the monument for an obliterated corner be reestablished by double or single proportionate measurement?

(A) double
(B) single
(C) neither—an obliterated corner is not lost
(D) single proportionment if it is on a township boundary; double if it is an interior corner

Solution: Proportionate measurement is a technique of last resort, used to reestablish the monumentation of lost corners. An obliterated monument is not considered lost; it can be located beyond a reasonable doubt by acceptable evidence.

Answer is (C)

68. A found closing corner monument, stamped AM, is found off the line upon which the closing was intended. Which statement best describes the likely circumstances?

(A) The found monument, being off of the intended line, was stamped AM, meaning *amended monument*.

(B) The found monument will help describe the closing line, but not its terminus.

(C) A new monument should have been established at the intersection of the closing line and the line intended to have been closed upon.

(D) A, B, and C are true.

Solution: Amended monuments are not only used to rectify off-line closing corners, but are also frequently found in such situations. The off-line monument, after being stamped AM, is connected by course and distance to the newly established monument at the true intersection of the closing line and the line closed upon.

Answer is (D)

69. What is the function of a location monument, sometimes designated by the abbreviation USLM?

(A) Where no public land corners are available within 2 mi, a USLM is established.

(B) The function of a USLM is to provide control for isolated special surveys or mineral surveys.

(C) A USLM is usually located on a prominent topographic feature.

(D) A, B, and C are true.

Solution: Location monuments were frequently established in areas where mineral claims or other bona fide private claims existed before the rectangular survey of the public lands in an area. A location monument in such a situation provided a common reference to which all the claims in an area could be tied. Later, the public lands survey would tie into the USLM and locate all the claims.

Answer is (D)

70. A meander corner is established at which of the following points?

(A) every point where a standard, township, or section line intersects the mean high water mark at the bank of a navigable body of water

(B) every point where an aliquot line intersects the mean high water mark at the bank of a meanderable body of water

(C) every point where a township line intersects the mean high water mark at the bank of a meanderable body of water

(D) only on banks of lakes 50 ac or larger

Solution: Meander lines are not established as legal boundaries, but rather as a method of setting off certain bodies of water from public lands. Meander lines are intended to follow the mean high water line along meanderable bodies of water. Meander corners are only established at the points where the meander line is intersected by standard, township, and section lines.

Answer is (A)

71. Which statement best defines the difference between a bearing tree and a line tree?

(A) A bearing tree is a corner accessory and a line tree is not.

(B) A bearing tree is blazed and a line tree is hacked.

(C) A line tree is intersected by a boundary line, while a bearing tree is scribed with the description of the section in which it stands.

(D) A, B, and C are true.

Solution: Trees have been marked since the inception of the Public Land Surveying System. Today, a line tree is used to perpetuate a surveyed line, while a bearing tree is considered a corner accessory. In both cases, the trees are marked to preserve healthy growth, in an attempt to ensure their future integrity.

Answer is (D)

72. Original monuments in the Public Land Surveying System are held in high regard. Which statement correctly defines their standing?

(A) Verified undisturbed original monuments in the public land system are, by law, coincident with the corners they were established to represent, except off-line closing corners.

(B) Verified undisturbed original monuments in the public land system are to remain fixed unless technical errors are discovered in the survey by which they were established. If such technical errors are discovered, the monuments may be moved to their correct location by any licensed land surveyor.

(C) The bearings and distances shown on official township plats are considered inferior evidence of the position of corners in the public land system when contradicted by verified original corner monumentation.

(D) Both A and C are true.

Solution: Once the returns from an original public lands survey have been officially accepted by the Bureau of Land Management, the monumentation set during that survey is inviolable.

Answer is (D)

73. Under current instructions, what is a regulation monument for public lands surveys established by the Bureau of Land Management?

(A) a #5 reinforcing bar with a plastic cap

(B) 4-in × 4-in × 36-in wooden post

(C) $3\frac{1}{2}$-in o.d. alloyed iron pipe, 28-in to 30-in long, with a brass cap

(D) $2\frac{1}{2}$-in o.d. alloyed iron pipe, 28-in to 30-in long, with a brass cap

Solution: Current Bureau of Land Management instructions specify a zinc-coated $2\frac{1}{2}$-in iron alloy pipe, split at its base, with a brass cap. A $3\frac{1}{4}$-in diameter brass tablet with a $3\frac{1}{2}$-in stem is also an official monument for use in rock and concrete. When authorized, durable native stone of an appropriate shape and a volume of at least 1000 cu in may be used in lieu of an iron post.

Answer is (D)

74. An original stone monument from a public lands survey of 1891 is found to have two notches cut into its eastern edge and three notches cut into its southern edge. What corner has been found?

(A) the SE corner of Section 10
(B) the SW corner of Section 14
(C) the NW corner of Section 3
(D) the NE corner of Section 28

Solution: The instructions for early public land surveys specified that the edges of stone monuments face the cardinal directions. These edges were then notched to indicate the number of miles that the monument stood from the governing eastern and southern township boundaries.

Answer is (B)

75. Under current instructions, which statement does not correctly define a difference between a bearing tree and a line tree?

(A) Bearing trees are corner accessories; line trees are not.
(B) Line trees are marked with horizontal hacks; bearing trees are scribed.
(C) Bearing trees are recorded in the field notes; line trees are not.
(D) Line trees are sometimes known also as *sight trees* or *station trees;* bearing trees are not.

Solution: The marking of trees has been part of the procedure of public lands surveying since its beginning. However, the nature and significance of the marks has changed over the years. Under the current instructions, trees that are intersected by a surveyed line are marked with two parallel horizontal hacks on the sides facing the line and are known as *line trees*. Bearing trees are corner accessories. They are generally within 3 chains of the corner. Normally, one bearing tree is selected in each section when trees of sufficient size and maturity are available. Each bearing tree is scribed with the letters BT and identification appropriate to the section in which it stands.

Both line trees and bearing trees should be recorded in the field notes of a survey.

Answer is (C)

76. A surveyor is retracing a section line that was blazed through a forested area following current instructions. The surveyor finds a tree with two blazes that are quite close together. The blazes are on the side of the tree facing the surveyor. Where is the section line in relation to the surveyor as he stands facing the blazes on the tree?

(A) The section line lies in front of the surveyor, approximately 3 chains distant.

(B) The section line lies in front of the surveyor, approximately 50 links distant.

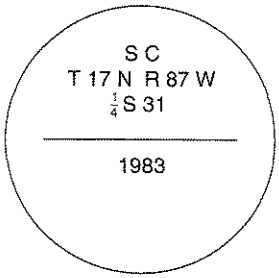
(C) The section line lies behind the surveyor, a foot or two distant.

(D) The section line lies behind the surveyor, approximately 50 links distant.

Solution: Trees are blazed within 50 links of a section line. Two blazes are made facing the line; the closer the blazes stand together on the tree, the farther the tree stands from the line.

Answer is (D)

77. If found on a regulation BLM brass cap monument, the stamping shown in the illustration would be evidence of which of the following corners?



(A) the SW corner of Section 17 on a standard parallel

(B) the SW corner of Section 31, T17N R87W

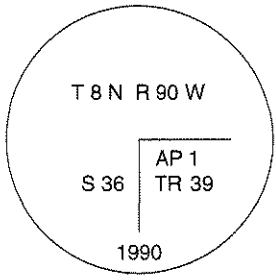
(C) the S$\frac{1}{4}$ corner of Section 31, T17N R87W, found on a standard parallel

(D) the S$\frac{1}{4}$ corner of Section 17

Solution: The abbreviation SC indicates that the monument stands at a standard corner, which must lie on a standard parallel.

Answer is (C)

78. If found on a regulation BLM brass cap monument, the stamping shown in the illustration would be evidence of which of the following corners?

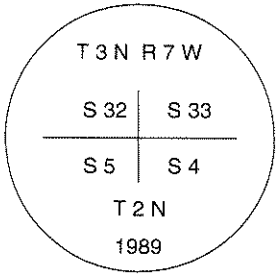


(A) a witness point for Section 36, T8N, R90W
(B) an auxiliary point for a land transfer in Section 36, T8N R90W
(C) the angle point number 1 for Tract 39 in Section 36, T8N R90W
(D) a witness corner for a desert land entry corner in Section 36, T8N R90W

Solution: The abbreviation AP stands for *angle point*. The abbreviation TR stands for *tract*. The tract is inside Section 36, T8N R90W.

Answer is (C)

79. If found on a regulation BLM brass cap monument, the stamping shown in the illustration would be evidence of which of the following corners?



(A) a section corner common to four sections on a standard parallel
(B) a section corner common to four sections that is not on a standard parallel
(C) a section corner on a township line
(D) both B and C

Solution: The absence of the abbreviations SC, for *standard corner*, or CC, for *closing corner*, indicates that the corner is probably not on a correction line. Also, Township 2 north is not likely to have a standard parallel as its northern boundary.

Answer is (D)

80. Which statement does not correctly represent a criterion for the placement of a witness corner monument?

(A) The preferred location of a witness corner is on one of the surveyed lines that leads to the true corner.
(B) A witness corner monument is established near a corner when the true corner falls in a position that makes monumentation and subsequent use impractical.
(C) The preferred distance between a witness corner and the true corner is 10 chains or less when the witness corner can be placed on a surveyed line that leads to the true corner.
(D) The 1973 *Manual of Surveying Instructions* calls for two witness corners to be placed near inaccessible true corners.

Solution: The 1973 *Manual* calls for one witness corner near inaccessible true corners, not two. The bearing and distance from the witness corner to the true corner are recorded in the field notes, and the witness corner is shown on the plat.

Answer is (D)

81. Which of the following abbreviations might be found on a corner accessory that was established near a section corner that fell in an unimproved roadway where neither bearing trees nor nearby bearing objects were available?

(A) AMC
(B) RM
(C) BT
(D) TR

Solution: When more common corner accessories are not available and the monument at a corner must be buried or is otherwise likely to be destroyed, a reference monument is set. A true corner that falls within a roadway may have two or four reference monuments. Reference monuments are described in field notes, but are not usually indicated on the official plat of the township. The abbreviation for a reference monument is RM.

Answer is (B)

82. A stone monument 20 in wide and 6 in thick is found projecting approximately 15 in from the ground. It has an "X" chiseled into the top and "USLM 34" is chiseled into the side. The monument was set in approximately 1906. Nearby rocks also have Xs and the letters "BR" chiseled into them. What sort of point does the monument represent and what is its purpose?

(A) It serves the same purpose as the U.S. Mineral Monuments formerly did.
(B) No corners of the public land system were available within 2 mi of the vicinity of a mineral survey when this USLM was established as a reference for the mineral surveys of the area.
(C) The monument was set by a U.S. Mineral Surveyor.
(D) A, B, and C are true.

Solution: The U.S. Location Monuments, formerly known as the U.S. Mineral Monuments, were set in mining districts as permanent references for mineral surveys before the Public Land Surveying System corners were available. The monuments were established by the U.S. Mineral Surveyors, and their latitudes and longitudes recorded.

Answer is (D)

83. Which monument would not be evidenced by an accessory under normal circumstances?

(A) a witness corner
(B) a closing corner
(C) a reference monument
(D) a meander corner

Solution: Accessories found in the vicinity of a reference monument refer to the position of the true corner rather than the reference monument, which is itself a corner accessory. It is not likely that other corner accessories will be found near a reference monument, since it is normally established in an area where other accessories are unavailable.

Answer is (C)

84. An old set of original field notes describes a cairn as a corner accessory. What is a cairn?

(A) a charred wooden post
(B) a tree species
(C) a mound of stone
(D) a pit

Solution: A cairn is a mound of stone. A mound of stone may still be used as a corner accessory per current instructions.

Answer is (C)

CORNER RECORDS

85. Several of the states covered by the Public Land Surveying System have enacted corner record legislation. Which of the following correctly defines a common objective of such legislation?

(A) The filing of corner records provides a mechanism by which original public lands survey corner monuments may be replaced with monuments established at the correct positions determined by private land surveyors.

(B) The filing of corner records helps to ensure the perpetuation of found original public lands survey corner monuments, and makes the evidence by which lost or obliterated monuments are reestablished a matter of public record.

(C) The filing of corner records provides a mechanism by which a private land surveyor may make monumentation of the aliquot corner of a section a matter of public record.

(D) Both B and C are true.

Solution: After public lands have been properly monumented, the official plats and field notes have been prepared, and the lands have passed into private ownership, the work of the BLM and its surveyors has been completed. The privately owned sectionalized lands then fall within the venue of the private land surveyors and under the jurisdiction of the state laws. These laws recognize the inviolability of the original public lands survey and its corner monumentation, almost without exception.

However, the original corner monuments may become obliterated or lost over time; many of the aliquot corners may not have been monumented at all by the original surveyors. These difficulties have been addressed in many states by corner recordation laws, which help to perpetuate privately set aliquot corner monuments, found original controlling monumentation, and lost or obliterated original corner monumentation.

Answer is (D)

INTERPRETATION OF FIELD NOTES

The following two problems refer to the following transcription of field notes.

Run north between Sections 25 and 26

20.40 Creek 6 links wide course N46°W, asc. SW slope.

38.16 Trail to Johnson's Mill, bears east and west.

40.00 Set a granite stone, $6 \times 12 \times 15$ inches in a mound of stone marked $\frac{1}{4}$ on west face for the $\frac{1}{4}$ corner of Sections 25 and 26 from which a fir, 16 inches diam., bears N84°E 48 links, marked $\frac{1}{4}$ S 25 BT, a hemlock 11 inches diam., bears N79°W, 88 links, marked $\frac{1}{4}$ S 26 BT.

68.40 Top of spur slopes N30°W, desc. north slope

80.00 Set a basalt stone $4 \times 16 \times 16$ inches in a mound of stone marked with 1 notch on the east and 2 notches on the south, for the corner of Sections 23, 24, 25, and 26. From the stone: a cedar, 13 inches diam., bears N25°E, 76 links marked T4N R3E S24 BT, a pine, 19 inches diam., bears S62°E, 16 links marked T4N R3E S25 BT, an oak, 26 inches diam., bears S10°W, 98 links marked T4N R3E S26 BT, an alder, 11 inches diam., bears N38°W, 19 links marked on the bark T4N R3E S23 BT.

Run north between Sections 23 and 24.

16.34 Sun Creek 46 links wide, courses N16°W, asc. steep south slope

39.34 Top of asc., enter meadow edges ...

86. What is the distance along the section line from the trail from Johnson's Mill to Sun Creek?

(A) 58.18 chains
(B) 16.34 chains
(C) 38.16 chains
(D) 21.82 chains

Solution: The distances recorded in the field notes of a public lands survey are cumulative. The distance from Johnson's Mill to the SE corner of Section 23 is 41.84 chains.

$$\begin{array}{r} 80.00 \text{ chains} \\ \underline{-38.16 \text{ chains}} \\ 41.84 \text{ chains} \end{array}$$

The distance from the SE corner of Section 23 to Sun Creek is 16.34 chains. The sum of the two distances is the distance from Johnson's Mill to Sun Creek.

$$\begin{array}{r} 41.84 \text{ chains} \\ \underline{+16.34 \text{ chains}} \\ 58.18 \text{ chains} \end{array}$$

Answer is (A)

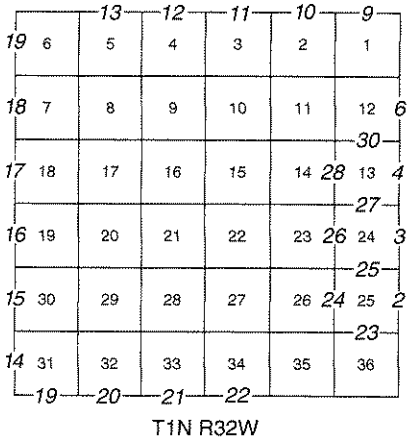
87. Which of the following terms describes the notation given as the distance to the trail to Johnson's Mill?

(A) crossing notes
(B) feature reports
(C) cultural certifications
(D) passing calls

Solution: The notations of cultural or topographic features crossed by a surveyed line are generally known as *passing calls*. Passing calls are frequently useful in recovering obliterated corners.

Answer is (D)

88. The following illustration is representative of a page found in most field notes of the public lands system. What is the significance of the illustration and the numbers shown on the township and section lines?



(A) It is an index, and the numbers refer to the consecutively numbered special instructions that were issued by the Bureau of Land Management to govern the survey.

(B) It is a diary, and the numbers refer to the days of the month upon which the section lines were surveyed.

(C) It is an index, and the numbers match the consecutive labels attached to the random lines recorded in the notes themselves.

(D) It is an index, and the numbers indicate the page of the notebook on which the measurement of the line is recorded.

Solution: An index diagram appears near the front of each field book. It is a convenient reference by which the notes of a particular survey line may be quickly found.

Answer is (D)

89. Which of the following governmental entities is the official custodian of the original field notes and plats of the completed public land surveys?

(A) All of the official public lands records are under the exclusive jurisdiction of the individual states.

(B) All of the official public lands records are under the exclusive jurisdiction of the director of the Bureau of Land Management in Washington, D.C.

(C) The official public lands records for all of the states east of the Mississippi River have been transferred to the states. All other records remain under the jurisdiction of the Bureau of Land Management.

(D) Where public lands surveys have been substantially completed, the official public lands records have been transferred to the states, though duplicates are on file in Washington, D.C. The remaining records are under the jurisdiction of the Bureau of Land Management.

Solution: As provided by federal law, the public lands survey records of states in which the survey is substantially complete have been transferred to the states' authority. These include all of the states east of the Mississippi, plus North Dakota, South Dakota, Nebraska, Kansas, Arkansas, and Louisiana.

Answer is (D)

90. Which of the following statements is not true of the official field note records of the public lands surveys that are prepared under the instructions in the 1973 *Manual of Surveying Instructions*?

(A) Field notes are typewritten transcriptions of the information recorded in field tablets.

(B) Random lines with fallings are recorded in the field notes.

(C) A full description of all of the monuments, both found and set, must be included in the field notes of a new survey.

(D) A general description of the soil, land, and forest along surveyed lines must be summarized at the conclusion of each mile completed.

Solution: The random lines and fallings are not recorded in the final transcribed field notes of a public lands survey. These trial lines, whose fallings are the basis of the calculation of true lines, were once included in the field notes of the public lands surveys, but are no longer.

Answer is (B)

THE ESTABLISHMENT OF ALIQUOT CORNERS

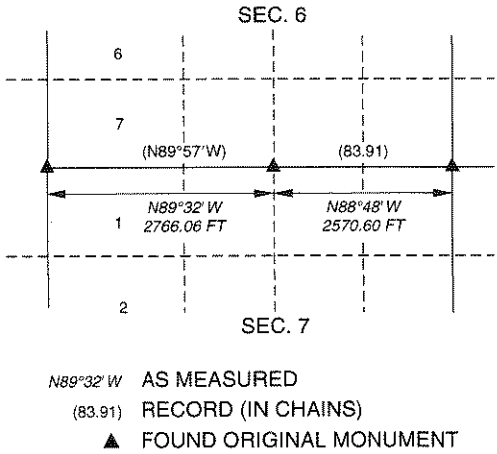
91. A private surveyor finds that his remeasurement along a section line between found original quarter corner and section corner monuments differs from that of the official record. The surveyor's objective is to establish an initial sixteenth corner monument along the line between the two found original corners. Which of the following techniques is the surveyor likely to find helpful in resolving the difference between the measurements?

(A) protraction
(B) double centering
(C) triangulation
(D) proportionate measurement

Solution: Proportionate measurement, also known as proration, is the technique of accommodating the principle that says that original record measurements cannot be disturbed. An even distribution of the discovered excess or deficiency is the objective of a proportionate measurement. The new values given to several parts of the line will bear the same relation to the record lengths as the new measurement of the whole line bears to the record of the whole line.

Answer is (D)

The following three problems refer to the illustration shown.



92. The NE, NW, and N$\frac{1}{4}$ corners of Section 7 are found original corner monuments. The official township plat indicates that the record bearing and distance across the north line of Section 7 is N89°57′W 83.91 chains. A private land surveyor intends to set the NE corner of Lot 1. The surveyor has measured N89°32′W 2766.06 ft between the N$\frac{1}{4}$ and the NW corners of Section 7. What distance should the surveyor measure west along the section line from the N$\frac{1}{4}$ corner to set the NE corner of Lot 1?

(A) 1259.88 ft
(B) 1320.00 ft
(C) 1383.03 ft
(D) 1506.18 ft

Solution: The first step in solving this problem is to find the record distance from the N$\frac{1}{4}$ to the NE corner of Lot 1. Since Section 7 is a closing section, the record distance 83.91 is not divided in half at the N$\frac{1}{4}$ corner. The record distance from the NE corner of Section 7 to its N$\frac{1}{4}$ corner is 40.00 chains. The remainder, 43.91 chains, is the record distance from the N$\frac{1}{4}$ corner of Section 7 to its NW corner. This 43.91 chains is not divided in half at the NE corner of Lot 1. The record distance from the N$\frac{1}{4}$ corner to the NE corner of Lot 1 is 20.00 chains; the remainder, 23.91 chains, is the record distance along the northern boundary of Lot 1.

The proportion of the record distance between the N$\frac{1}{4}$ corner of Section 7 and its NW corner, 43.91 chains, to the record distance from the N$\frac{1}{4}$ to the NE corner of Lot 1, 20.00 chains, must be preserved.

$$20.00 \text{ chains} = 1320.00 \text{ ft}$$
$$43.91 \text{ chains} = 2898.06 \text{ ft}$$
$$\frac{1320.00 \text{ ft}}{2898.06 \text{ ft}} = \frac{x}{2766.06 \text{ ft}}$$
$$3{,}651{,}199.20 \text{ ft} = 2898.06x$$
$$x = \frac{3{,}651{,}199.20 \text{ ft}}{2898.06}$$
$$= 1259.88 \text{ ft}$$

Answer is (A)

93. The NE, NW, and N$\frac{1}{4}$ corners of Section 7 are found original corner monuments. The official township plat indicates that the record bearing and distance across the north line of Section 7 is N89°57′W 83.91 chains. A private land surveyor intends to set the NE corner of NW$\frac{1}{4}$NE$\frac{1}{4}$ of Section 7. The surveyor has measured N88°48′W 2570.60 ft between the NE and the N$\frac{1}{4}$ corners of Section 7. What distance should the surveyor measure west along the section line from the NE corner to set the NE corner of NW$\frac{1}{4}$NE$\frac{1}{4}$?

(A) 1320.00 ft
(B) 1250.98 ft
(C) 1285.30 ft
(D) 1349.20 ft

Solution: The proportion of the record distance between the NE corner of Section 7 and its N$\frac{1}{4}$ corner, 40.00 chains, to the record distance from the NE corner of Section 7 to the NE corner of NW$\frac{1}{4}$NE$\frac{1}{4}$, 20.00 chains, must be preserved.

$$20.00 \text{ chains} = 1320.00 \text{ ft}$$
$$40.00 \text{ chains} = 2640.00 \text{ ft}$$

$$\frac{1320.00 \text{ ft}}{2640.00 \text{ ft}} = \frac{x}{2570.60 \text{ ft}}$$

$$3{,}393{,}192.00 \text{ ft} = 2640.00x$$

$$x = \frac{3{,}393{,}192.00 \text{ ft}}{2640.00}$$

$$= 1285.30 \text{ ft}$$

Answer is (C)

94. Under normal circumstances and by current instructions, would the original surveyor of Section 7 have established a monument at the SW corner of Lot 1?

(A) Yes, the lot corners along a township boundary are set during the course of the original survey of the township boundary.

(B) The SW corner of Lot 1 would have been monumented during the course of the original survey only if the western boundary of Section 7 was a guide meridian.

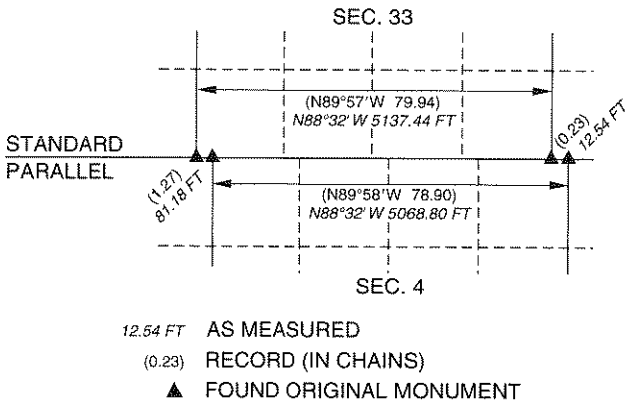
(C) No, neither the quarter-quarter corners nor the lot corner are monumented during the course of the original survey of a township, unless mandated by special instructions.

(D) The SW corner of Lot 1 would have been monumented during the course of the original survey only if it was a closing corner.

Solution: Unless otherwise instructed, the original surveyor does not monument the quarter-quarter corners or lot corners of a township.

Answer is (C)

The following two problems refer to the illustration shown.



95. Retracing a standard parallel, a surveyor discovers that the quarter corners were not established by the original surveyor. Using the record and measured information given, what distance should be measured along the standard parallel from the eastern standard corner to the northern quarter corner of Section 4?

(A) 2603.70 ft
(B) 2638.02 ft
(C) 2534.40 ft
(D) 2521.86 ft

Solution: The record distance between the closing corners at the NE and NW corners of Section 4 is 78.90 chains, or 5207.40 ft. The measured distance between the same two corners is 5068.80 ft. The record distance between the NE corner of Section 4 and the N$\frac{1}{4}$ corner is half the record distance between the NE and NW corners, or 2603.70 ft. The quarter corner monument should be placed on the standard parallel and halfway between the NE and NW corners of Section 4 by proportionate measure.

$$\frac{2603.70 \text{ ft}}{5207.40 \text{ ft}} = \frac{x}{5068.80 \text{ ft}}$$

$$13{,}197{,}634.56 \text{ ft} = 5207.40x$$

$$x = \frac{13{,}197{,}634.56 \text{ ft}}{5207.40}$$

$$= 2534.40 \text{ ft}$$

The distance from the NE corner to the N$\frac{1}{4}$ corner of Section 4 is 2534.40 ft. However, the eastern standard corner is the SE corner of Section 33, which lies 12.54 ft west of the NE corner of Section 4.

$$\begin{array}{r} 2534.40 \text{ ft} \\ \underline{-12.54 \text{ ft}} \\ 2521.86 \text{ ft} \end{array}$$

Answer is (D)

96. What distance would be measured between the N$\frac{1}{4}$ corner of Section 4 and the S$\frac{1}{4}$ corner of Section 33?

(A) 46.86 ft
(B) 68.64 ft
(C) 49.50 ft
(D) 55.12 ft

Solution: The record distance from the SE corner to the SW corner of Section 33 is 79.94 chains, or 5276.04 ft. The record distance from the SE corner to the S$\frac{1}{4}$ corner of the section is half that, or 2638.02 ft. The measured distance along the south line of Section 33 is 5137.44 ft; the S$\frac{1}{4}$ corner must be half that distance from the SE corner.

$$\left(\tfrac{1}{2}\right)(5137.44 \text{ ft}) = 2568.72 \text{ ft}$$

Similarly, the measured distance from the NE corner of Section 4 to its N$\frac{1}{4}$ corner is half the measured distance between the NE and NW corners.

$$\left(\tfrac{1}{2}\right)(5068.80 \text{ ft}) = 2534.40 \text{ ft}$$

The distance from the NE corner of Section 4 to its N$\frac{1}{4}$ corner must be reduced by 12.54 ft to find the distance from the SE corner of Section 33 to the N$\frac{1}{4}$ corner of Section 4.

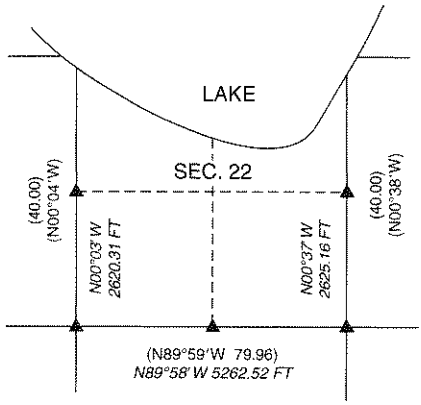
$$\begin{array}{r} 2534.40 \text{ ft} \\ \underline{-12.54 \text{ ft}} \\ 2521.86 \text{ ft} \end{array}$$

The distance between the N$\frac{1}{4}$ corner of Section 4 and the S$\frac{1}{4}$ corner of Section 33 can now be found.

$$\begin{array}{r} 2568.72 \text{ ft} \\ -2521.86 \text{ ft} \\ \hline 46.86 \text{ ft} \end{array}$$

Answer is (A)

The following two problems refer to the illustration shown.



N89°58' W 5262.52 FT NEW MEASUREMENT
(N89°59'W 79.96) RECORD (IN CHAINS)
▲ FOUND ORIGINAL MONUMENT

97. The illustration indicates the record dimensions and the measurements of a retracement of a fractional Section 22. The surveyor retracing the section wishes to monument the center of the section. What bearing should the surveyor use along the meridional centerline of the section?

(A) N00°20′12″W
(B) N00°00′00″W
(C) N00°19′24″W
(D) N00°20′00″W

Solution: Where opposite corresponding corners cannot be set in a fractional section, it is proper to run from the established corner as nearly parallel to the section lines as possible.

The procedure most often used to satisfy the spirit of the rule is to adopt a mean, or average, course from the established corner. In this case, the retracement surveyor has found the eastern section line to have a bearing of N00°37′W and the western section line to have a bearing of N00°03′W.

$$\frac{\text{N00°37′W} + \text{N00°03′W}}{2} = \text{N00°20′00″W}$$

Answer is (D)

98. The retracement surveyor wishes to set the NE corner of the SE$\frac{1}{4}$SE$\frac{1}{4}$, also known as the ES$\frac{1}{16}$. Which bearing and distance would represent the correct measurement from the SE corner of Section 22 to this quarter-quarter corner?

(A) N00°37′W, 1312.58 ft
(B) N00°38′W, 1320.00 ft
(C) N00°38′W, 1312.58 ft
(D) N00°37′W, 1320.00 ft

Solution: The subdivision of the SE$\frac{1}{4}$ of Section 22 includes the division of its boundaries by half. The ES$\frac{1}{16}$ corner is halfway between the SE corner and E$\frac{1}{4}$ corners of the section. Therefore, the record indicates a bearing of N00°38′W and a distance of 20.00 chains from the SE corner to the ES$\frac{1}{16}$. However, a retracement of the line indicates N00°37′W and 1312.58 ft.

$$\begin{aligned}
\frac{1320.00 \text{ ft}}{2640.00 \text{ ft}} &= \frac{x}{2625.16 \text{ ft}} \\
3{,}465{,}211.20 \text{ ft} &= 2640.00x \\
x &= \frac{3{,}465{,}211.20 \text{ ft}}{2640.00} \\
&= 1312.58 \text{ ft}
\end{aligned}$$

Answer is (A)

THE REESTABLISHMENT OF CORNERS

99. Which statement best defines the meaning of the term *obliterated corner* as used in the Public Land Surveying System?

(A) The monument of an obliterated corner remains in place and can be verified by other record evidence.

(B) The position of an obliterated corner cannot be recovered beyond a reasonable doubt.

(C) There are no traces of the monument at an obliterated corner, but its position can be recovered beyond a reasonable doubt.

(D) The position of an obliterated corner can only be restored by reference to one or more interdependent corners.

Solution: The monument is missing from an obliterated corner, as well as its accessories. However, the location has been perpetuated and can be located beyond a reasonable doubt without relying completely on reference to other, interdependent corners.

Answer is (C)

100. Which statement best defines the meaning of the term *existent corner* as used in the Public Land Surveying System?

(A) A corner should not be declared existent until every possible means has been unsuccessful in locating its true original position.

(B) No traces of the original monument or its accessories are found at the true position of an existent corner, nor can they be located by record evidence.

(C) It is a corner whose position is identifiable by the monument, accessories, or other physical or record evidence.

(D) It is a corner whose position can only be proven by reference to one or more other interdependent corners.

Solution: The evidence of the monument or its accessories may be present at an existent corner. An existent corner can sometimes be identified by reference to the description in field notes or by an acceptable supplemental survey record, physical evidence, or testimony.

Answer is (C)

101. Which description best defines the meaning of the term *lost corner* as it is used in the Public Land Surveying System?

(A) a corner whose position is indeterminable, except by reference to one or more interdependent corners

(B) a corner whose position is indeterminable, except by testimony of one or more witnesses who have dependable knowledge of its original location

(C) a corner whose position is indeterminable, except by reference to its accessories

(D) a corner whose position is indeterminable, except by reference to an acceptable supplemental survey record, physical evidence, or testimony

Solution: A corner is not lost until all efforts to locate its position by original marks, acceptable record evidence, and testimony are unsuccessful. A corner whose position can only be restored by reference to one or more interdependent corners is considered lost.

Answer is (A)

102. Which of the following statements is not true of the use of proportionate measurement for the restoration of corners in the Public Land Surveying System?

(A) It should be considered a procedure of last resort.

(B) The new survey of a line must be adjusted so that it will be in proportionate agreement with the original reported length. The original reported length is considered the true length of the line.

(C) The extension of any proportionate measurement beyond an original monument is incorrect. Proportionate measurement should only be used between original monuments.

(D) Proportionate measurement cannot be applied to restore lost closing corners along a standard parallel.

Solution: Double proportionate measurement is not applicable to the restoration of corners along a standard parallel, since the alignment of the standard line is more important than the section lines that close upon the parallel.

Answer is (D)

103. Which of the following lost corners would be restored by double proportionate measurement?

(A) a lost quarter corner
(B) a lost corner on a township boundary that is common to four sections
(C) a lost standard corner on a correction line
(D) a lost corner common to four townships

Solution: Double proportionate measurement is used to restore a lost corner between four known corners on lines of equal importance. A lost corner common to four interior sections or a lost corner common to four townships would be subject to double proportionate measurement.

Answer is (D)

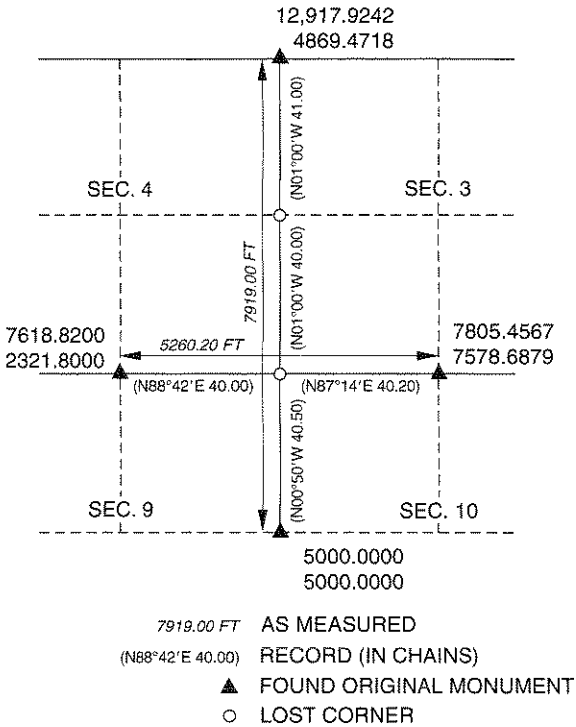
104. Which of the following lost corners would not be restored by single proportionate measurement?

(A) a lost quarter section corner on a township boundary
(B) a lost quarter section corner on a section line within a township
(C) a lost meander corner
(D) none of the above

Solution: Single proportionate measurement is applied to lost corners where one line has precedence over another.

Answer is (D)

The following three problems refer to the illustration shown.



105. The surveyor who retraced the section lines indicated in the illustration wishes to reestablish both of the lost corners on the meridional line between Sections 3 and 4. Which statement correctly describes the proper sequence of the work?

(A) First, the lost quarter section corner should be restored by single proportionment.

(B) First, the lost section corner should be restored by double proportionment.

(C) It makes no difference which of the two lost corners is restored first.

(D) First, the lost quarter section corner should be restored by double proportionment.

Solution: The lost section corner must be restored by double proportionate measurement before the lost quarter section corner is restored, using single proportionate measurement.

Answer is (B)

106. What are the coordinates of the restored section corner common to Sections 3, 4, 9, and 10? (In the following work, the first coordinate is the northing and the second is the easting.)

(A) 7711.9056
4956.4906

(B) 7639.3081
4943.6892

(C) 7675.6068
4950.0899

(D) 7703.9512
4932.7405

Solution: The lost section corner must be reestablished by double proportionate measurement. The first step is the establishment of a temporary stake at the proper proportional distance from the nearest found original corner to the north and the south on the meridional section line. The record indicates that the distance from the W$\frac{1}{4}$ corner of Section 10 to the NW corner of Section 10 is 40.50 chains, or 2673.00 ft.

$$(40.50 \text{ chains})(66.00 \text{ ft}) = 2673.00 \text{ ft}$$

The record also indicates that the distance from the SW corner of Section 3 to the NW corner of Section 3 is 81.00 chains, or 5346.00 ft.

$$\begin{array}{r} 40.00 \text{ chains} \\ \underline{+41.00 \text{ chains}} \\ 81.00 \text{ chains} \end{array}$$

$$(81.00 \text{ chains})(66.00 \text{ ft}) = 5346.00 \text{ ft}$$

The measured distance from the W$\frac{1}{4}$ corner of Section 10 to the NW corner of Section 3 is 7919.00 ft. The total record distance for the same line is 8019.00 ft.

$$\begin{array}{r} 2673.00 \text{ ft} \\ +5346.00 \text{ ft} \\ \hline 8019.00 \text{ ft} \end{array}$$

Apply single proportionate measurement to find the distance from the W$\frac{1}{4}$ corner of Section 10 to the temporary stake on the meridional line.

$$\begin{aligned} \text{Record:} \quad & \text{Measured:} \\ \frac{2673.00 \text{ ft}}{8019.00 \text{ ft}} &= \frac{x}{7919.00 \text{ ft}} \\ 21{,}267{,}487.00 \text{ ft} &= 8019.00x \\ x &= \frac{21{,}267{,}487.00 \text{ ft}}{8019.00 \text{ ft}} \\ &= 2639.67 \text{ ft} \end{aligned}$$

Check the calculation by finding the proportional distance from the NW corner of Section 3 to the temporary stake on the meridional line.

$$\begin{aligned} \text{Record:} \quad & \text{Measured:} \\ \frac{5346.00 \text{ ft}}{8019.00 \text{ ft}} &= \frac{x}{7919.00 \text{ ft}} \\ 42{,}334{,}974.00 \text{ ft} &= 8019.00x \\ x &= \frac{42{,}334{,}974.00 \text{ ft}}{8019.00 \text{ ft}} \\ &= 5279.33 \text{ ft} \end{aligned}$$

The sum of the two proportional distances calculated should equal the total measured distance.

$$\begin{array}{r} 5279.33 \text{ ft} \\ +2639.67 \text{ ft} \\ \hline 7919.00 \text{ ft} \end{array}$$

The azimuth of the line from the W$\frac{1}{4}$ corner of Section 10 to the NW corner of Section 3 is found by inversing the coordinates.

W$\frac{1}{4}$ corner of Section 10:		NW corner of Section 3:
5000.0000	$359°03'20''$	12,917.9242
5000.0000	7919.00 ft	4869.4718

Using this azimuth and the proportionate distance calculated, find the coordinates of the temporary stake on the meridional line.

W$\frac{1}{4}$ corner of Section 10:		Temporary stake on the meridional line:
5000.0000 5000.0000	$\frac{359°03'20''}{2639.67 \text{ ft}}$	7639.3081 4956.4906

Next, find the coordinates of the temporary stake on the latitudinal line.

The record distance from the N$\frac{1}{4}$ corner of Section 9 to the lost NE corner of Section 9 is 40.00 chains, or 2640.00 ft. The record distance from the N$\frac{1}{4}$ corner of Section 9 to the N$\frac{1}{4}$ corner of Section 10 is 80.20 chains, or 5293.20 ft. The new measurement of the same distance is 5260.20 ft.

$$\begin{aligned} \text{Record:} \quad & \text{Measured:} \\ \frac{2640.00 \text{ ft}}{5293.20 \text{ ft}} &= \frac{x}{5260.20 \text{ ft}} \\ 13{,}886{,}928.00 \text{ ft} &= 5293.20x \\ x &= \frac{13{,}886{,}928.00 \text{ ft}}{5293.20} \\ &= 2623.54 \text{ ft} \end{aligned}$$

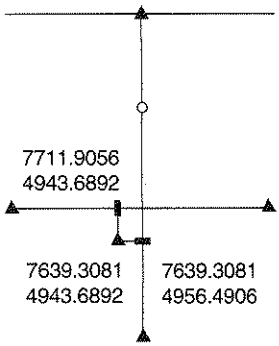
This is the proportionate distance from the N$\frac{1}{4}$ corner of Section 9 to the temporary stake on the latitudinal line. To find the azimuth of the same line, inverse between the coordinates given for the N$\frac{1}{4}$ corner of Section 9 and the N$\frac{1}{4}$ of Section 10.

N$\frac{1}{4}$ corner Section 9:		N$\frac{1}{4}$ corner Section 10:
7618.8200 2321.8000	$\frac{87°58'00''}{5260.20 \text{ ft}}$	7805.4567 7578.6879

Using this azimuth and the proportionate distance calculated, find the coordinates of the temporary stake on the latitudinal line.

N$\frac{1}{4}$ corner Section 9:		Temporary stake on the latitudinal line:
7618.8200 2321.8000	$\frac{87°58'00''}{2623.54 \text{ ft}}$	7711.9056 4943.6892

By intersecting cardinal lines passing through the two temporary stakes, as shown in the following illustration, the coordinates of the reestablished corner common to Sections 3, 4, 9, and 10 can be found.



Answer is (B)

107. Referring to the illustration, what are the coordinates of the restored quarter-section corner common to Sections 3 and 4?

(A) 10,263.1875
4943.6892

(B) 10,307.6741
4982.2100

(C) 10,296.8641
4910.5398

(D) 10,246.0321
4907.0386

Solution: First, inverse between the coordinates given for the NW corner of Section 3 and the reestablished corner at the SW corner of Section 3.

Reestablished SW corner of Section 3:		NW corner of Section 3:
12,917.9242	179°11′40″	7639.3081
4869.4718	5279.1378 ft	4943.6892

The record distance for the same line is 81.00 chains, or 5346.00 ft.

$$\begin{array}{r} 40.00 \text{ chains} \\ \underline{+41.00 \text{ chains}} \\ 81.00 \text{ chains} \end{array}$$

$$(81.00 \text{ chains})(66.00 \text{ ft}) = 5346.00 \text{ ft}$$

Use single proportionate measurement to find the required distance along the west line of Section 3.

$$\begin{array}{rcl} \text{Record:} & & \text{Measured:} \\ \dfrac{2640.00 \text{ ft}}{5346.00 \text{ ft}} & = & \dfrac{x}{5279.1378 \text{ ft}} \\ 13{,}936{,}923.79 \text{ ft} & = & 5346.00x \\ x & = & \dfrac{13{,}936{,}923.79 \text{ ft}}{5346.00} \\ & = & 2606.9816 \text{ ft} \end{array}$$

The proportional distance from the SW corner of Section 3 to the W$\frac{1}{4}$ corner is 2606.9816. It is advisable to check the work by proportioning the remainder of the line as well.

$$\begin{array}{rcl} \text{Record:} & & \text{Measured:} \\ \dfrac{2706.00 \text{ ft}}{5346.00 \text{ ft}} & = & \dfrac{x}{5279.1378 \text{ ft}} \\ 14{,}285{,}346.89 \text{ ft} & = & 5346.00x \\ x & = & \dfrac{14{,}285{,}346.89 \text{ ft}}{5346.00} \\ & = & 2672.1562 \text{ ft} \end{array}$$

The sum of the two proportional distances along the west line of Section 3 should equal the total measured length.

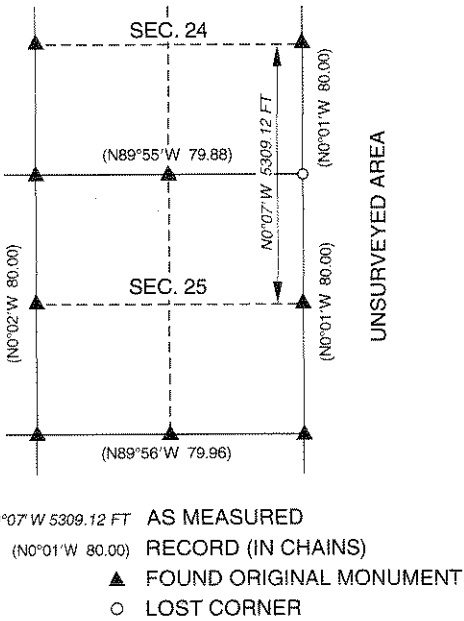
$$\begin{array}{r} 2606.9816 \text{ ft} \\ \underline{+2672.1562 \text{ ft}} \\ 5279.1378 \text{ ft} \end{array}$$

Finally, the coordinates of the reestablished W$\frac{1}{4}$ corner of Section 3 can be found by proceeding north along the west line of the section from its SW corner, using the azimuth and proportional distance found.

SW corner of Section 3:		W$\frac{1}{4}$ corner of Section 3:
7639.3081	$359°11'40''$	10,246.0321
4943.6892	2606.9816 ft	4907.0386

Answer is (D)

108. In the following illustration, the NE corner of Section 25 is lost. Which bearings and distances from the adjacent found original corners to the position of the restored NE corner of Section 25 are correct?



(A) from the E$\frac{1}{4}$ corner of Section 25, N0°01′W 2640.00 ft to the restored NE corner of Section 25

(B) from the N$\frac{1}{4}$ corner of Section 25, N89°55′W 2636.04 ft to the restored NE corner of Section 25

(C) from the E$\frac{1}{4}$ corner of Section 25, N0°07′W 2654.56 ft to the restored NE corner of Section 25

(D) both A and B

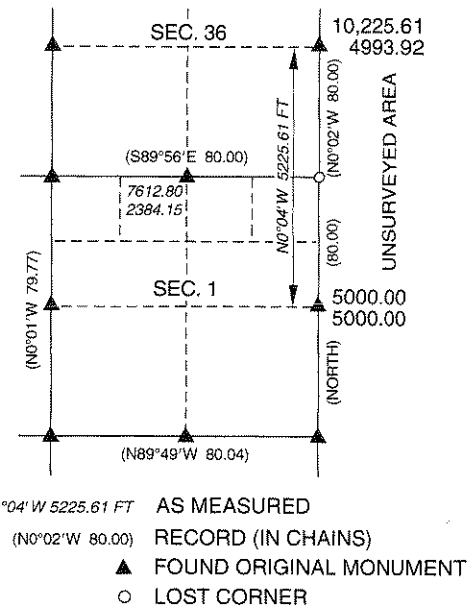
Solution: The lost corner is on a township boundary and must be restored by single proportionate measurement from the nearest identified regular corners to the north and south. The record shows that the township line is straight; the retracement of the line indicates that its bearing is N0°07′W. The record distance from the missing point to the E$\frac{1}{4}$ corner of Section 25 is 40.00 chains.

The distance to the E$\frac{1}{4}$ corner of Section ~~26~~ 24 is also 40.00 chains. Therefore, the missing point lies halfway between these two known points.

$$\frac{5309.12 \text{ ft}}{2} = 2654.56 \text{ ft}$$

Answer is (C)

The following two problems refer to the illustration shown.



109. Which of the following procedures would be used to restore the lost NE corner of Section 1?

(A) single proportionate measurement
(B) double proportionate measurement
(C) one-point control
(D) three-point control

Solution: Under normal circumstances, a lost township corner such as that illustrated would be restored by double proportionate measurement. However, the latitudinal township line has not been established east of the lost corner and double proportionment is not possible. The correct procedure in such a case is to use three-point control. The meridional line is divided proportionally and a temporary stake set, as in double proportionment. However, the record bearing and distance is used to establish the second temporary stake along the latitudinal line. Cardinal lines through the temporary stakes intersect at the restored corner.

Answer is (D)

110. Which of the following coordinates represent the position of the properly restored corner at the NE corner of Section 1?

(A) 7609.73
5024.15

(B) 7612.80
4996.96

(C) 7702.71
5001.43

(D) 7612.80
5024.15

Solution: The first step in restoring the NE corner of Section 1 is to set a temporary stake on the meridional township line at the proper proportional distance. The record indicates that the line between the E$\frac{1}{4}$ of Section 1 and the E$\frac{1}{4}$ corner of Section 36 is 80.00 chains, or 5280.00 ft. However, the retracement of the line shows it to be 5225.61 ft. The record establishes that the proper proportion between the line E$\frac{1}{4}$ to NE of Section 1 and the line from NE Section 1 to E$\frac{1}{4}$ of Section 36 is half and half.

$$\begin{aligned} \text{Record:} \quad & \text{Measured:} \\ \frac{2640.00 \text{ ft}}{5280.00 \text{ ft}} &= \frac{x}{5225.61 \text{ ft}} \\ 13{,}795{,}610.40 \text{ ft} &= 5280.00x \\ x &= \frac{13{,}795{,}610.40 \text{ ft}}{5280.00} \\ &= 2612.80 \text{ ft} \end{aligned}$$

The distance from the E$\frac{1}{4}$ corner of Section 1 to the temporary stake on the meridional line is 2612.80 ft, or one-half of 5225.61 ft. The bearing from the

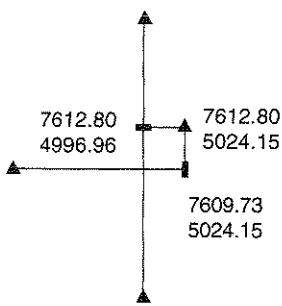
$E\frac{1}{4}$ corner of Section 1 and this temporary stake is the bearing measured in the retracement of the line N00°04′W.

$E\frac{1}{4}$ corner of Section 1:		Temporary stake on the meridional line:
5000.00 5000.00	N00°04′W 2612.80 ft	7612.80 ~~5024.15~~ 4996.96

Next, the temporary stake on the latitudinal line is placed at the record bearing and distance from the $N\frac{1}{4}$ corner of Section 1.

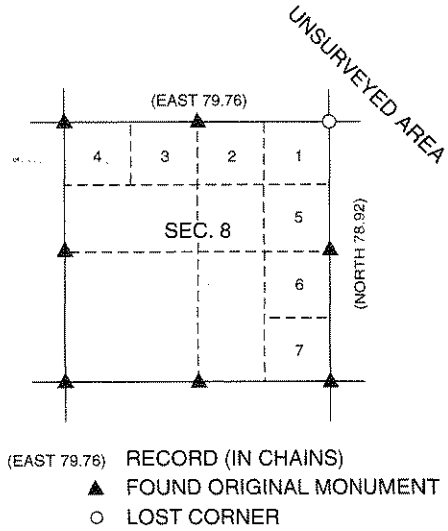
$N\frac{1}{4}$ corner Section 1:		Temporary stake on the meridional line:
7612.80 2384.15	S89°56′E 2640.00 ft	7609.73 5024.15

Cardinal lines through these temporary stakes intersect at the correct location of the restored corner.



Answer is (D)

The following two problems refer to the illustration shown.



111. Which of the following procedures would be used to restore the lost NE corner of Section 8?

(A) one-point control
(B) two-point control
(C) single proportionate measurement
(D) double proportionate measurement

Solution: The lost section corner cannot be restored by either single or double proportionment, since the lines north and east of the lost corner have not been established. The record bearing and distance to the E$\frac{1}{4}$ and N$\frac{1}{4}$ corners of the section must be used to restore the corner. Temporary stakes should be set north of the E$\frac{1}{4}$ corner and east of the N$\frac{1}{4}$ corner at the record bearings and distances. Cardinal lines established through these stakes will intersect at the restored section corner. The procedure is known as *two-point control.*

Answer is (B)

112. To the nearest tenth of a chain, what would be the distances from the E$\frac{1}{4}$ and N$\frac{1}{4}$ corners of Section 8 to the restored NE corner of the section?

(A) The distance from the N$\frac{1}{4}$ corner would be 39.9 chains. The distance from the E$\frac{1}{4}$ corner would be 39.5 chains.

(B) The distance from the N$\frac{1}{4}$ corner would be 39.9 chains. The distance from the E$\frac{1}{4}$ corner would be 38.9 chains.

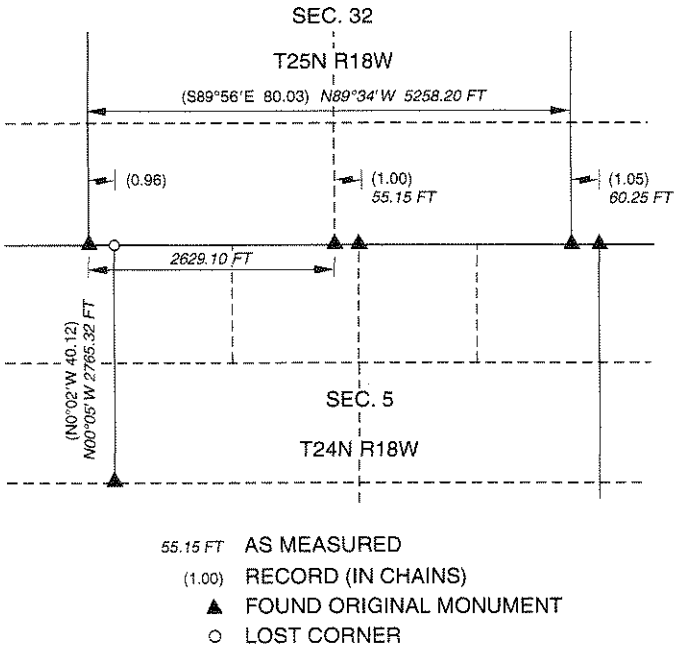
(C) The distance from the N$\frac{1}{4}$ corner would be 39.8 chains. The distance from the E$\frac{1}{4}$ corner would be 39.5 chains.

(D) The distance from the N$\frac{1}{4}$ corner would be 39.8 chains. The distance from the E$\frac{1}{4}$ corner would be 38.9 chains.

Solution: The illustration indicates that Section 8 has been lotted on the north and east. Therefore, the distance from the NW corner to the N$\frac{1}{4}$ corner of Section 8 is 40.00 chains, and the distance from the N$\frac{1}{4}$ corner to the N-E$\frac{1}{16}$ is 20.00 chains. The closing distance to the NE corner of the section is therefore 19.76 chains from the west, and 18.92 chains from the south.

Answer is (D)

The following two problems refer to the illustration shown.



113. Which of the following procedures would be used to restore the lost NW corner of Section 5?

(A) one-point control
(B) two-point control
(C) single proportionate measurement
(D) double proportionate measurement

Solution: The closing corner at the NW corner of Section 5 must be restored by single proportionate measurement. Standard parallels and township exteriors have precedence over subdivisional lines.

Answer is (C)

114. To the nearest tenth of a foot, what distance would be measured from the properly restored closing corner at the NW corner of Section 5 to the SW corner of Section 32?

(A) 63.4 ft
(B) 56.7 ft
(C) 57.7 ft
(D) 63.1 ft

Solution: The measured distance from the S$\frac{1}{4}$ corner to the SW corner of Section 32 is 2629.10 ft. The record distance for the same line is half of 80.03 chains, or 2640.99 ft. The proportionate distance should be calculated according to the falling recorded to the nearest regular corner: 0.96 chains, or 63.36 ft.

$$\begin{aligned} \text{Record:} \quad & \text{Measured:} \\ \frac{63.36 \text{ ft}}{2640.99 \text{ ft}} &= \frac{x}{2629.10 \text{ ft}} \\ 166{,}579.78 \text{ ft} &= 2640.99x \\ x &= \frac{166{,}579.78 \text{ ft}}{2640.99} \\ &= 63.1 \text{ ft} \end{aligned}$$

Answer is (D)

RESURVEYS

115. Which of the following statements best defines the meaning of the term *dependent resurvey*?

(A) It is a resurvey that is based upon existing corners and acceptable control points from the original survey. It includes the restoration of lost corners by proportionate measurement.

(B) It is a resurvey that overrides the original survey, establishes new section lines, and also protects bona fide private land holdings in segregated tracts.

(C) It is a resurvey that is made to find the direction and distance between previously established corners, and to rehabilitate recovered corners. It does not include the restoration of lost corners.

(D) It is a resurvey that is made with the intention of finding the lengths of the lines previously established.

Solution: A dependent resurvey is done on the foundation created by the original survey; it does not supersede the original survey. A dependent resurvey does not include the segregation of private holdings into tracts that were previously aliquot parts. In law, the aliquot parts established by the original survey are reestablished unchanged by a dependent resurvey.

Answer is (A)

116. Which of the following terms has the same meaning as dependent resurvey in the Public Land Surveying System?

(A) retracement
(B) remeasurement
(C) independent resurvey
(D) relocation

Solution: Retracement refers to a survey of public lands that is done to determine the length and direction of the lines previously established; it does not include the restoration of lost corners. Remeasurement refers only to the measurement of the lengths of lines. Relocation generally refers to the movement of a mining claim within public lands.

Answer is (D)

117. Which of the following statements is not true of an independent resurvey in the Public Land Surveying System?

(A) It is a resurvey that supersedes the original survey of a portion of the public lands.
(B) Private bona fide rights are protected in the creation of tracts by metes and bounds surveys.
(C) The outboundary, which is the limit of the land to be independently surveyed, is established using the rules of a dependent resurvey.
(D) The lines established in an independent resurvey close upon the corner established by the original survey.

Solution: An independent resurvey is not done on the foundation created by the original survey. It supersedes the original survey. An independent resurvey includes the segregation of private holdings into tracts that were previously aliquot parts. The aliquot parts established by the original survey are not reestablished by the independent resurvey.

Answer is (D)

118. The numbering of tracts that are to be segregated by a metes and bounds survey within an independent resurvey always begins with what number?

(A) 1
(B) 100
(C) 25
(D) 37

Solution: The first segregated tract is numbered 37 to avoid confusion with the original sections, which are numbered 1 to 36.

Answer is (D)

119. Which of the following abbreviations is likely to appear on a monument at the corner of a tract in an independently resurveyed township?

(A) LM
(B) WP
(C) BO
(D) AP

Solution: The corners of the tracts in an independently resurveyed township are known as angle points, and the abbreviation AP appears on the monuments of those corners. The angle points are numbered consecutively. Number 1 is usually assigned to the NE corner of the tract, with the numbers increasing counterclockwise around the tract. A corner of each tract is tied to a corner of the resurvey.

Answer is (D)

120. Which of the following statements is not true of a dependent resurvey in the Public Land Surveying System?

(A) Local points of control may be accepted, and such points are afforded the same authority as an identified original corner.

(B) The records kept of a dependent resurvey must conform to the specifications for original surveys.

(C) Monuments found remaining in place from the original survey are destroyed unless the point is needed to control a bona fide private claim.

(D) Both A and C are untrue.

Solution: Old monuments are destroyed during an independent resurvey, since such a survey is intended to supersede the original work. However, such found monuments are definitely not destroyed during the work of a dependent resurvey.

Answer is (C)

LOTTING

121. Which of the following best describes the meaning of the word *lot* as it is used in the Public Land Surveying System?

(A) a quarter-quarter section that adjoins the northern or western boundary of a township and bears a lot number

(B) any nonaliquot portion of a section

(C) a quarter-quarter section that contains less than 5 ac

(D) subdivisions of a section that are not described as aliquot parts but bear a lot number

Solution: Subdivisions of sections that are not aliquot parts and that bear assigned lot numbers on the official township plat are considered lots in the Public Land Surveying System.

Answer is (D)

122. Which of the following conditions would likely not result in the creation of lots in a section?

(A) the sections along the exterior of a township absorbing the excess or deficiency of measurement accumulated within the township
(B) sections adjoining a meandered lake, pond, or river
(C) the closure of lines of the rectangular survey upon the boundaries of a reservation or national park
(D) a retracement of the north line of a section that previously contained no lots reveals a discrepancy with the original measurement of 3 min in direction and 4 links in length

Solution: A retracement of a section does not lead to the the creation of lots in that section.

Answer is (D)

123. Which of the following statements is true of an elongated section?

(A) An elongated section contains no quarter-quarter sections.
(B) An elongated section contains more than two tiers of lots.
(C) An elongated section exceeds 85 chains in its length, width, or both.
(D) Both B and C are true.

Solution: An elongated section may contain regular quarter-quarter sections. For example, a section may be elongated north of its meridional quarter line and still contain a regular SE and SW quarter.

Answer is (D)

124. Which of the following statements correctly describes the system of lotting used in a section that has been the subject of an independent resurvey?

(A) The designation of the parts of such a section as lots is determined in the same way as in any other section that is subject to lotting.

(B) A portion of a resurveyed section may be given a lot number if it would otherwise have the same aliquot description it had before the resurvey.

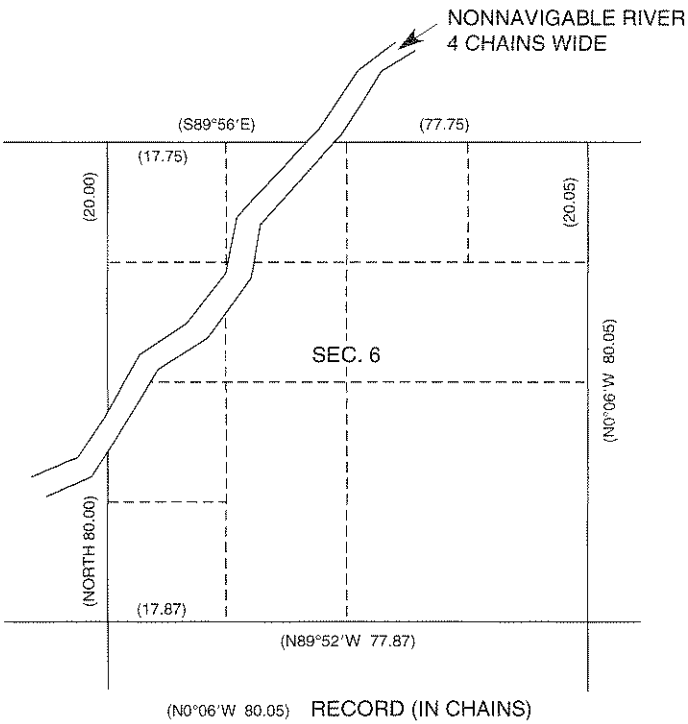
(C) Lot numbers begin with the next number higher than the highest lot number previously used in the section.

(D) Both B and C are true.

Solution: The creation of lots in an area covered by an independent resurvey is somewhat different than in areas not covered by such a resurvey. The rules governing the lots' positions and numbering are designed to avoid any possible confusion in description with the parts of the original survey.

Answer is (D)

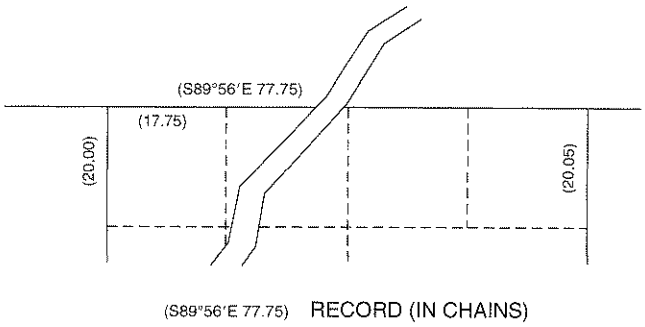
The following two problems refer to the illustration shown.



125. What number would be assigned to the most northwesterly lot in the section?

(A) 1
(B) 2
(C) 4
(D) 5

Solution: The numbering of lots in a section is similar to the numbering of sections in a township. It begins at the most northeasterly lot and increases consecutively to the west in the first tier. The next tier of lots south is numbered from west to east, the next is numbered from east to west, and the most southerly tier of lots is numbered from east to west.



Answer is (D)

126. How many lots are there in the section?

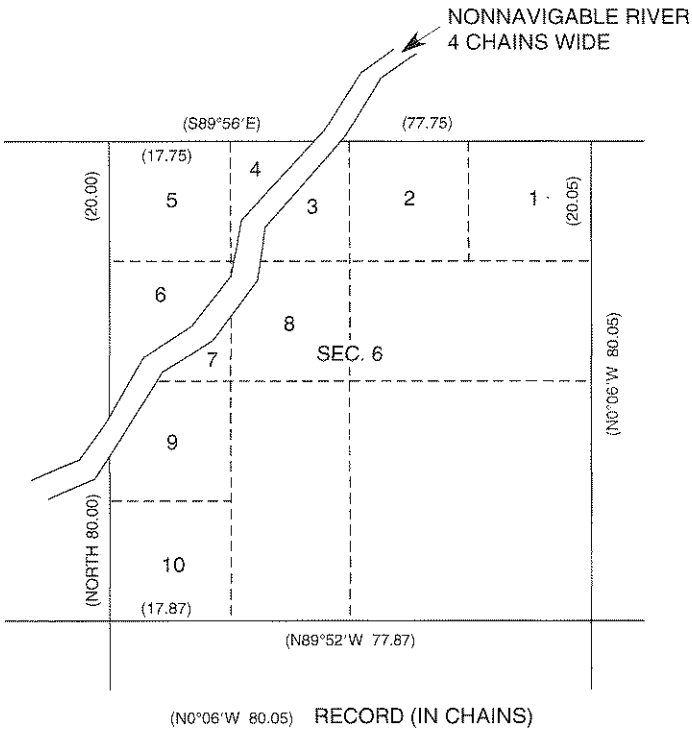
(A) 7 lots
(B) 6 lots
(C) 10 lots
(D) 8 lots

Solution: The river that crosses the section is not navigable, but it is more than 3 chains wide and, therefore, meanderable. The parts of the section that have

been invaded by the river are considered lots, along with the lots created by the excess and deficiency of measurement from the rest of the township. There are 10 lots in the section.

Answer is (C)

The following four problems refer to the illustration shown.



127. Which of the following statements gives the most likely reason that the southern boundary of Lot 6 is not a portion of the latitudinal centerline of the section?

(A) An error was made by the draftsman who protracted the section.
(B) The southern boundary of Lot 6 is a portion of the centerline of the section, regardless of what appears on the plat.
(C) The portion of land created between the meander line, the centerline, and the western boundary of the section was considered too small to be a lot.
(D) The area labeled as Lot 6 is actually not a lot.

Solution: The most probable explanation is that an area was created that was considered too small or irregular to constitute a lot. The protraction of the section was likely arranged to add the small area to another adjoining lot.

Answer is (C)

128. What is the implied record distance from the southeastern corner of Lot 7 to the W$\frac{1}{4}$ corner of Section 6?

(A) 20.00 chains
(B) 17.81 chains
(C) 17.77 chains
(D) 17.75 chains

Solution: The required length is the average of the parenthetical distances given along the northern and southern section lines for the western range of lots. The western section line has a continuous bearing, and the meridional lengths of each lot are equal at 20.00 chains. Therefore, the average of the parentheticals at each end will be correct.

$$\begin{array}{r} 17.75 \text{ chains} \\ \underline{+17.87 \text{ chains}} \\ 35.62 \text{ chains} \end{array}$$

$$\frac{35.62 \text{ chains}}{2} = 17.81 \text{ chains}$$

Answer is (B)

129. What are the implied record lengths of the eastern and southern boundaries of Lot 5?

(A) The eastern boundary is 20.00 chains. The southern boundary is 17.75 chains.

(B) The eastern boundary is 20.10 chains. The southern boundary is 17.90 chains.

(C) The eastern boundary is 20.0125 chains. The southern boundary is 17.78 chains.

(D) The eastern boundary is 20.01 chains. The southern boundary is 17.78 chains.

Solution: One method of solving this problem is to find the change in the parenthetical distances per the unit of length of the adjacent section line. For example, looking first at the western range of lots, the difference between the given parenthetical distances is 0.12 chains. Dividing the difference between these parentheticals by the total length of the western section line yields the rate of change.

$$\begin{aligned} \text{southernmost parenthetical} &= 17.87 \text{ chains} \\ \text{northernmost parenthetical} &= \underline{-17.75 \text{ chains}} \\ & \ 0.12 \text{ chains} \end{aligned}$$

$$\frac{0.12 \text{ chains}}{80.00 \text{ chains}} = 0.0015$$

This ratio can be used to determine the difference between the consecutive parenthetical distances along the western range of lots. Multiply this ratio by the meridional length of Lot 5 to find the record length.

$$(20.00 \text{ chains})(0.0015) = 0.03 \text{ chains}$$

The difference in the record lengths of the southern line and the northern line of Lot 5 is 0.03 chains.

$$\begin{aligned} \text{length of the northern line of Lot 5} &= 17.75 \text{ chains} \\ & \underline{+0.03 \text{ chains}} \\ \text{length of the southern line of Lot 5} &= 17.78 \text{ chains} \end{aligned}$$

The same approach can be used to determine the record length of the eastern line of Lot 5.

$$\begin{aligned} \text{easternmost parenthetical} &= 20.05 \text{ chains} \\ \text{westernmost parenthetical} &= \underline{-20.00 \text{ chains}} \\ &\ \ 0.05 \text{ chains} \end{aligned}$$

$$\frac{0.05 \text{ chains}}{77.75 \text{ chains}} = 0.000643$$

Multiply this ratio by the length of the northern line of Lot 5.

$$(17.75 \text{ chains})(0.000643) = 0.01 \text{ chains}$$

$$\begin{aligned} \text{length of the western line of Lot 5} &= 20.00 \text{ chains} \\ &\ \underline{+0.01 \text{ chains}} \\ \text{length of the eastern line of Lot 5} &= 20.01 \text{ chains} \end{aligned}$$

Answer is (D)

130. On the west side, the ownership of the patent holder of Lot 3 most likely ends at which of the following lines?

(A) the meander line
(B) the mean high water line
(C) the low water line
(D) the thread of the stream

Solution: The meander line was not established to define the limits of ownership of patent holders, but to indicate the sinuosities of the banks of the stream. Since the stream is not navigable, it is most likely that the thread of the stream is the boundary of the owner of Lot 3.

Answer is (D)

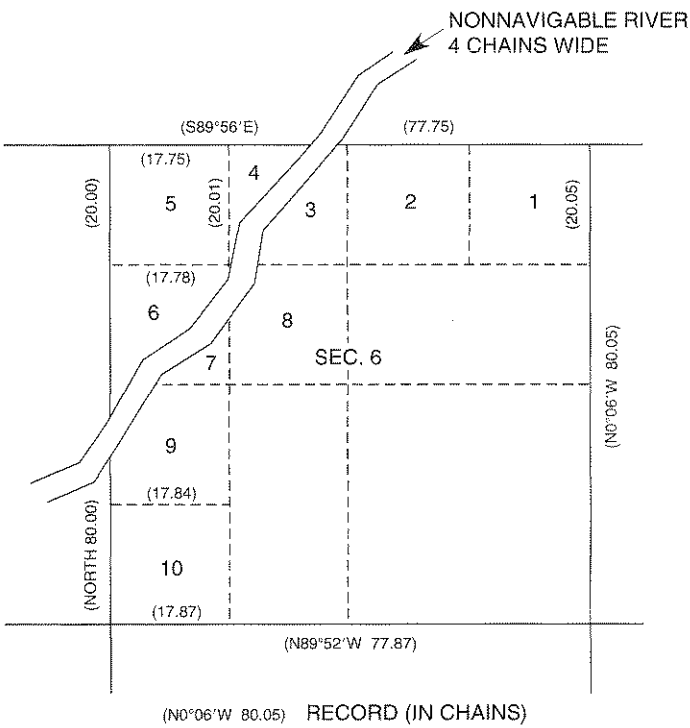
131. In modern practice, the creation of poorly shaped or odd-sized lots is avoided. Which of the following statements correctly defines the desirable limits on the areas of lots?

(A) A lot with a width of 20 chains should contain between 10 and 50 ac.
(B) A lot along an irregular boundary should contain between 5 and 45 ac.
(C) A lot along an irregular boundary should contain between 10 and 60 ac.
(D) Both A and B are true.

Solution: Extreme lengths and narrow widths are avoided in the creation of lots, and a lot should not extend beyond a section or quarter line. The lots along meander lines and other broken boundaries are generally not smaller than 5 ac or larger than 45 ac in modern practice. The lots with a full, normal width are generally not smaller than 10 ac or larger than 50 ac.

Answer is (D)

The following two problems refer to the illustration shown.



132. What is the record area of Lot 10?

(A) 40.00 ac
(B) 35.71 ac
(C) 35.60 ac
(D) 36.21 ac

Solution: One convenient aspect of the use of the chain as a unit of measurement is that 10 sq chains are equal to 1 ac. The area of Lot 10 computed by coordinates is approximately 35.68 ac. However, it is unlikely that the record area would be so computed, since the boundaries of the lot are regular. One simple method that might be employed is to multiply the averages of the opposite lot boundaries.

$$\frac{17.84 \text{ chains} + 17.87 \text{ chains}}{2} = 17.855 \text{ chains}$$

$$\frac{20.00 \text{ chains} + 20.00 \text{ chains}}{2} = 20.00 \text{ chains}$$

$$(20.00 \text{ chains})(17.855 \text{ chains}) = 357.10 \text{ sq chains}$$

$$(357.10 \text{ sq chains})\left(\frac{1 \text{ ac}}{10 \text{ sq chains}}\right) = 35.71 \text{ ac}$$

An even simpler method of finding the acreage in cases where the lot is 20.00 chains wide is to add the lengths of the opposite lot boundaries.

$$\begin{array}{r} 17.84 \text{ chains} \\ \underline{+17.87 \text{ chains}} \\ 35.71 \text{ ac} \end{array}$$

Answer is (B)

133. What is the record area of Lot 5?

(A) 35.54 ac
(B) 35.59 ac
(C) 35.65 ac
(D) 35.71 ac

Solution: To find the area, average the lengths of the opposite sides of the lot and multiply the results.

$$\frac{17.78 \text{ chains} + 17.75 \text{ chains}}{2} = 17.765 \text{ chains}$$

$$\frac{20.00 \text{ chains} + 20.01 \text{ chains}}{2} = 20.005 \text{ chains}$$

$$(20.005 \text{ chains})(17.765 \text{ chains}) = 355.39 \text{ sq chains}$$

$$(355.39 \text{ sq chains})\left(\frac{1 \text{ ac}}{10 \text{ sq chains}}\right) = 35.54 \text{ ac}$$

Answer is (A)

134. A lot with regular boundaries has a northern and southern boundary of 20.00 chains. The record area of the lot is 39.72 ac. The eastern boundary of the lot has a record length of 19.84 chains. What is the record length of the western boundary of the lot?

(A) 20.00 chains
(B) 19.84 chains
(C) 19.88 chains
(D) 19.90 chains

Solution: The area of a lot with a width of 20.00 chains and regular boundaries is equal to the sum of the parenthetical distances. Since the area and one parenthetical distance are known, the other parenthetical distance can be derived.

$$\text{parenthetical distance} = \begin{array}{r} 39.72 \text{ ac} \\ \underline{-19.84 \text{ chains}} \\ 19.88 \text{ chains} \end{array}$$

Answer is (C)

EXCEPTIONS AND DISCREPANCIES IN THE PUBLIC RECORD

135. Which of the following statements is not correct regarding corners and lines that have been accepted as the official record by the U.S. Secretary of the Interior?

(A) They are established as the proper corners and boundary lines, and their positions and lengths cannot be changed by any private land survey.

(B) They are considered to enclose the exact quantity of land expressed on the official township plat.

(C) The lines of subdivision that have not actually been run by the official government survey should be established by running straight lines between opposite corresponding corners.

(D) The Bureau of Land Management has the authority to conduct an official resurvey of any township.

Solution: The Bureau of Land Management has the authority to conduct an official government resurvey when more than half of the area of a township remains in the public domain. Otherwise, the resurvey must be requested by at least three-quarters of the entrymen, and the estimated cost of the resurvey deposited.

Answer is (D)

136. In general, meander lines are not legal boundaries. Which statement correctly describes a circumstance that could cause a meander line to become a legal boundary?

(A) Lotting within a section is shown abutting a meander line on the official township plat.

(B) The meander line occurs along a river that is part of the boundary between two states.

(C) The meander line occurs on an island.

(D) The meander line was established where no meanderable body of water ever existed.

Solution: When land has been omitted from the public lands survey by a fraudu lent meander line or one that is in gross error, the meander line may be made fixed boundary, as between surveyed and unsurveyed land.

Answer is (D)

RIGHTS OF POSSESSION IN PUBLIC LANDS

137. In general, unwritten property rights will not ripen against the federa government, but exceptions can arise. Which of the following statements correctl represents an exception to the general rule?

(A) The Bureau of Land Management is authorized to issue a patent to individuals who claim adverse possession of public domain lands under th Color of Title Act.

(B) In some cases, land may be acquired by adverse possession from federa agencies when such land has been previously patented into private ownership and has since been reacquired by the federal government.

(C) In some cases, land held by federal agencies never was part of the public domain; some of these lands may be acquired by adverse possession.

(D) A, B, and C are true.

Solution: Reserved public lands may be acquired by adverse possession through satisfaction of strict conditions under the Color of Title Act, 43 C.F.R. § 2540. The patents, which fall under two classifications, cannot be for more than 160 ac. Where valuable improvements or cultivation have been maintained for more than 20 years or taxes paid since January 1, 1901, under color of title, in good faith, peacefully, and by a claimant, his ancestors, or grantors, a patent may be issued.

Answer is (D)

MEANDERING

138. Which of the following best defines the meaning of the term *meander line*?

(A) a traverse run along the mean low water line of a natural body of water
(B) a line run to indicate the sinuosities of the bank or shore line of a permanent body of water
(C) a line used to determine the approximate area of land in fractional sections after the segregation of the water area
(D) both B and C

Solution: The meander line along a permanent natural body of water should follow the mean high water line.

Answer is (D)

139. Which of the following best defines the meaning of the term *meander corner*?

(A) a corner established at each deflection point of a meander line
(B) a corner established at the intersection of each aliquot line with a meander line
(C) a randomly placed corner used to control a meander line
(D) a corner established at the intersection of a meander line with a standard, township, or section line

Solution: The individual courses of the traverses that constitute a meander line are not monumented. Meander corners are established only where the meandering intersects standard, township, or section lines.

Answer is (D)

140. Which of the following intersections would be properly monumented by a special meander corner?

(A) where a surveyed aliquot line intersects a meander line
(B) where a calculated section centerline intersects a meander line
(C) where a meander line encloses swamp or overflow lands
(D) both A and B

Solution: Where aliquot lines have been surveyed through a section, their intersection with a meander line is monumented by a special meander corner. A meander line around a lake larger than 50 ac and enclosed entirely within a section will frequently include special meander corners. The position of special meander corners in such circumstances is calculated along the section centerline.

Answer is (D)

141. Which of the following conditions would be accommodated by the establishment of an auxiliary meander corner?

(A) a meandered lake entirely enclosed within a quarter section
(B) a meanderable body of water in an unsurveyed area of a township
(C) an island too small to subdivide that lies entirely in one section
(D) both A and C

Solution: An auxiliary meander corner is established to provide a method of connecting a meander line to a regular section corner. A small island that is meandered, but otherwise not tied to a section line, and a meandered lake not crossed by any quarter line both can be handled by the establishment of an auxiliary meander corner.

Answer is (D)

142. Under rules set down in the 1973 *Manual of Surveying Instructions*, which of the following should be meandered?

(A) rivers and streams having a width of 3 chains or more, whether navigable or not
(B) lakes covering 40 ac or more
(C) swamp lands
(D) islands above the mean high water line that have arisen after the state in which they stand was admitted into the U.S.

Solution: Lakes of 50 ac and more should be meandered under current instructions. Swamp and overflow lands should not be meandered. Islands that have been in existence since before a state was admitted to the U.S. should be meandered, but not islands that have arisen after statehood. All navigable rivers, streams, and bayous should be meandered.

Answer is (A)

MINERAL SURVEYS

143. Which of the following acts contains provisions that are still in effect regarding the regulation of mining claims on public land?

(A) the Act of July 25, 1866
(B) the Act of March 3, 1853
(C) the Act of February 5, 1805
(D) the Act of May 10, 1872

Solution: The Act of May 10, 1872 is also known as the Mining Act and R.S. 2331 (30 U.S.C. 35). The provisions of this law still guide the survey of mineral claims on public lands.

Answer is (D)

144. Which type of surveyor is authorized to survey a mineral claim on public land in preparation for a patent?

(A) a licensed land surveyor
(B) a Bureau of Land Management cadastral surveyor
(C) a U.S. deputy mineral land surveyor
(D) a licensed geologist

Solution: A survey for the patent of a mineral claim on public land must be conducted by a U.S. deputy mineral land surveyor. The Bureau of Land Management will provide a list of approved mineral surveyors to any claimant requiring such services.

Answer is (C)

145. In surveying a mineral lode claim for patent, a U.S. deputy land surveyor must make sure that the claim conforms to which of the following specifications?

(A) an aliquot part of a section
(B) 800 ft by 2000 ft
(C) 20 chains by 40 chains
(D) 600 ft by 1500 ft

Solution: The Mining Act of 1872 provides that a mineral lode claim may not exceed 600 ft by 1500 ft. It need not be aliquot.

Answer is (D)

146. A claimant may wish to make the end lines of a mineral lode claim parallel to one another. Why is such a configuration advantageous to the claimant?

(A) It guarantees that the claim contains the maximum possible area on the surface.
(B) It is required by the 1872 Mining Act.
(C) It allows for the maximum extralateral rights.
(D) The claimant may not otherwise amend the lines when the lode claim is surveyed for patent.

Solution: The sidelines of a mineral claim are often parallel with the outcrop of the vein on the surface, and the end lines are parallel to one another. This arrangement allows the miner to follow the dip of the ore into the earth, even when the pursuit carries the work beyond the boundaries of the claim. This is known as an extralateral right. The Mining Act of 1872 specifically calls for the end lines to be parallel.

Answer is (C)

147. What sort of mineral deposits are appropriate for inclusion in a placer claim?

(A) alluvial deposits
(B) veins in place
(C) glacial deposits
(D) both A and C

Solution: A claim can be made upon public lands for the purpose of placer mining. A placer claim may include minerals such as particles in sands or gravel, but not veins in place, which are the subjects of lode claims.

Answer is (D)

148. What provisions currently govern the configuration of a placer claim?

(A) A placer claim must be no more than 600 ft by 1500 ft.
(B) A placer claim must include no more than 300 ft on each side of the thread of a stream.
(C) A placer claim must enclose no more than 160 ac.
(D) A placer claim must conform as nearly as is practical to the rectangular subdivisions of the Public Land Surveying System.

Solution: Placer claims, unlike lode claims, must be surveyed in conformance with the aliquot lines of the Public Land Surveying System. Latitude is given regarding this specification in areas where placer claims are filed in advance of the rectangular survey.

Answer is (D)

149. What is the maximum area of public land an individual may hold as a placer claim under current provisions?

(A) 40 ac
(B) 20 ac
(C) 80 ac
(D) 10 ac

Solution: Under the provisions of the Mining Act of 1872, each claimant may hold 20 ac as a placer claim.

Answer is (B)

150. What is the maximum area of public land an individual may claim as a mill site?

(A) 20 ac
(B) $2\frac{1}{2}$ ac
(C) 40 ac
(D) 5 ac

Solution: Five acres may be included within a mill site for the processing of minerals; however, the mill site must be located on lands that do not contain valuable minerals.

Answer is (D)

151. Which of the following provisions is not required of a mineral survey for the patent of a lode claim?

(A) The lengths of the lines are to be returned in feet rather than chains.
(B) A true astronomic bearing must be determined on one of the lines of the survey.
(C) The mineral surveyor may move the claim stakes in, but not out, to bring it within statutory limits.
(D) The mineral survey must show the patented lode claim in the same position as recorded in the original location certificate.

Solution: A U.S. deputy mineral surveyor may survey a claim for patent that does not conform to the original mineral location certificate. The claimant may have moved the claim and obtained an amended location certificate any number of times.

Answer is (D)